# 0.1 What Is Market Risk?

## 0.1.1 Introduction

Learning Objectives

Understand market risk; distinguish instruments from factors; use a risk factor space and conditional scenario models to connect market states, portfolio values, and future P&L distributions; use an option case to link pricing, revaluation, P&L attribution, VaR and ES with explicit treatment of time decay; and understand the scenario/valuation interface, the two uses of Monte Carlo, and nested simulation and acceleration for exotic portfolios.
**Chapter Keywords:**

- **Risk Factors**, **Risk Factor Space**, **Conditional Scenario Distribution**.
- **Correlation Matrix**, **Cholesky Decomposition**, **Monte Carlo Scenario Generation**.
- **Historical Simulation**, **GARCH / Copula / HMM**, **Transformer / Diffusion**.
- **Black–Scholes Pricing**, **Full Revaluation**, **Delta / Gamma / Vega / Theta / Rho**.
- **Risk Engine**, **Real-World / Risk-Neutral Measures**, **Nested Monte Carlo**, **Pricing Surrogate**.
- **Frozen Valuation Date**, **Time Decay**, **P&L Attribution**, **Loss Distribution**, **VaR and ES**.

### Uncertainty in Market Prices

Asset prices in financial markets respond continually to macroeconomic conditions, supply and demand, investor behavior, and liquidity conditions. For an institution holding financial assets or trading positions[1], this uncertainty affects asset valuations and therefore portfolio value. Even when position sizes remain unchanged, changing market conditions can produce gains or losses.

Definition 0.1: Market Risk

**Market risk** is the risk of a decline in the value of a financial instrument or portfolio caused by adverse movements in market prices or market variables.

The phrase *potential loss* emphasizes that risk is forward-looking. The absence of a loss today does not imply the absence of market risk tomorrow. A price increase may also generate a loss on a short position. Whether a market movement is adverse therefore depends on the direction of the position and the instrument's valuation structure, rather than on whether a price rises or falls in isolation.

[1] **Position**: The quantity and direction of a holding in an asset or contract. Long positions generally benefit from a rise in the underlying price and short positions from a fall, while derivative exposures depend on contractual structure.

## From Market States to Portfolio Value

The fundamental relationship can be written as:

$$V = f(X),$$

Here, $V$ denotes the value of a financial asset or portfolio, $X$ is the **market risk factor vector**[2], and $f$ is a **valuation function**[3] mapping a market state to a value. This relationship separates the problem into two connected components: how market states evolve, and how much the portfolio is worth in a given state.

To make time, positions, and parameters explicit, write:

$$V_t = f(t, X_t; \theta), \qquad X_t = \begin{bmatrix} x_{1,t} \\ x_{2,t} \\ \vdots \\ x_{n,t} \end{bmatrix} \in \mathcal{X} \subseteq \mathbb{R}^n.$$

$x_{i,t}$ is the state of risk factor $i$ at time $t$, and $n$ is the number of factors. The symbol $\theta$ represents positions, contractual terms, and model specifications held fixed for the risk calculation. It does not imply that every model parameter must be fixed: a parameter whose uncertainty is modeled as a risk source belongs in $X_t$. For a portfolio of multiple assets, the valuation function usually aggregates position values expressed in a chosen reporting currency.

## Risk Factor Space

The **risk factor space**[4] $\mathcal{X}$ is the set of admissible market states under the selected factor representation. $X_t$ is today's state; $\mathcal{X}$ specifies where current and future states may lie. Its definition establishes coordinates, units, and validity constraints, giving scenario generation and valuation a common description of market states.

For the simplified equity call in this chapter, take $X_t = (S_t, \sigma_t, r_t)^\mathsf{T}$ and choose:

$$\mathcal{X}_{\text{call}} = (0, \infty) \times (0, \infty) \times \mathbb{R}.$$

Spot and implied volatility are positive, while rates may be negative. Strike and contractual expiry remain fixed terms, and remaining maturity follows from the valuation date. This space applies to the example's factor representation, rather than every asset: the WTI example below requires negative prices to be admissible for that contract, so a log-price model cannot be imposed universally.

Curves and volatility surfaces are commonly represented by nodes or parameters; $\mathcal{X} \subseteq \mathbb{R}^n$ then denotes the state space of that finite-dimensional representation. Admissible coordinates must also permit reconstruction of valid discount curves and option prices, with the relevant no-arbitrage checks. An arbitrary collection of real numbers need not define a market that can

[2] **Risk factor**: A variable used to describe a market state that affects instrument value or the distribution of future P&L. It may be an observable price or a curve or parameter estimated from market quotes.

[3] **Valuation function**: A rule for calculating value given a market state, contractual terms, and positions, ranging from quantity times price to a complex derivative pricing model.

[4] **Risk factor space**: The set of admissible market states in chosen coordinates, including domain and curve/surface reconstruction constraints. It defines allowed states, not their probabilities.

be priced. Path-dependent contracts may additionally require historical fixings or accumulated states; these must enter the state representation or fixed contract information rather than being assumed recoverable from current spot alone.

The factor space specifies which states are admissible, the probability model specifies how likely they are, and valuation specifies what the portfolio is worth in them. These definitions must be consistent.

## Economic Meaning of Risk Factors

Risk factors are market variables that drive changes in valuation; financial instruments are the assets or contracts an institution holds. The distinction matters: a bond is an instrument, while the yield curve[5] and credit spread[6] affecting its value are risk factors. Factor selection determines which value changes the model can represent and which sources of risk it may omit.

The factor vector can be partitioned by common sources of risk as:

$$X_t = \begin{bmatrix} X_t^{\mathrm{IR}} \\ X_t^{\mathrm{FX}} \\ X_t^{\mathrm{EQ}} \\ X_t^{\mathrm{COM}} \\ X_t^{\mathrm{CS}} \\ X_t^{\mathrm{VOL}} \end{bmatrix}.$$

Each block may itself be a vector. The interest-rate block may contain rates at several maturities, while the volatility block may contain implied volatilities for different expiries and strike prices. The expression concatenates six factor categories; it does not restrict the model to six scalar factors.

Table 1: Risk factors, typical exposures and historical cases (sources linked).

| Factor | Meaning | Typical Exposure | Case |
|---|---|---|---|
| **IR**: Interest rates | Rates and yield curves | Bonds, interest-rate swaps | **US Treasury yields**; SVB, 2023: preceding rate increases reduced long-duration securities' values, creating substantial valuation losses. |
| **FX**: Foreign exchange | Currency exchange rates | Foreign-currency assets, FX forwards | **EUR/CHF**; Swiss franc shock, 2015: removal of the exchange-rate floor triggered a sharp EUR/CHF fall and losses on short-franc positions. |
| **EQ**: Equity | Stock and equity-index prices | Stocks, index futures | **S&P 500 index**; Black Monday, 1987: the equity-market collapse generated large losses on long stock and equity-index positions. |

[5] **Yield curve**: A curve relating yields to maturities under specified conventions. Valuation also distinguishes discount and forward curves, as well as zero-coupon yields and yields to maturity.

[6] **Credit spread**: The yield spread of a debt instrument over a selected benchmark, usually quoted in basis points; one basis point equals 0.01 percentage points. Spreads may reflect credit, liquidity, and risk premia. Spread-movement risk should be distinguished from an actual default event.

| Factor | Meaning | Typical Exposure | Case |
|---|---|---|---|
| **COM**: Commodities | Energy, agricultural and metal prices | Commodity futures and options | **WTI crude oil futures prices**; negative oil prices, April 2020: the May contract settled below zero, hurting nearby-contract longs and exposing delivery and storage constraints. |
| **CS**: Credit spreads | Yield spreads over a benchmark | Corporate bonds, credit derivatives | **US high-yield OAS**; global financial crisis, 2008: sharply widening credit spreads exerted downward pressure on credit-bond valuations. |
| **VOL**: Volatility | Implied or realized volatility[7] | Options, volatility derivatives | **VIX futures prices**; volatility shock, February 2018: surging volatility futures generated severe losses on inverse-volatility products. |

The cases identify observable market variables and illustrate how losses propagate. Events commonly combine multiple factors, leverage and liquidity; SVB also involved a deposit run. **OAS (option-adjusted spread)** measures a spread after adjusting for embedded-option effects under a valuation model. VIX-related products require revaluation using the futures or contracts they actually track. The negative WTI price episode also shows that commodity scenario models must permit negative prices for relevant contracts and check the applicability of transformations such as log returns.

These categories do not form a one-to-one classification of assets. A single instrument often has several exposures: a corporate bond depends on rates and credit spreads; a foreign-currency bond also introduces exchange-rate exposure; and an equity option typically depends on the underlying price, volatility, and interest rates. The direct valuation of an ordinary stock position generally depends on its price, while return volatility affects the distribution of its future profit and loss. This differs from an option whose value depends directly on implied volatility.

Factors may also move together. Under market stress, falling equity prices, widening credit spreads, and rising implied volatility may occur simultaneously. Portfolio risk modeling therefore requires both the behavior of individual factors and the **dependence structure**[8] linking them.

## Risk Factor Models: From State Space to Conditional Scenarios

After defining $\mathcal{X}$, the model must specify a **conditional distribution**[9] for future states. For a risk horizon $h > 0$, let $\mathcal{F}_t$ denote information available at $t$, including market history and estimated volatility or regime states. Scenario generation can then be written uniformly as:

$$X_{t+h}^{(j)} \sim \mathsf{P}_{t,h}(\cdot \mid \mathcal{F}_t), \qquad X_{t+h}^{(j)} \in \mathcal{X}.$$

[7] **Volatility**: A measure of variability, commonly the standard deviation of returns over a specified timescale. Realized volatility is estimated from observed price changes; implied volatility is inferred from option prices under a specified pricing model. The two are distinct.

[8] **Dependence structure**: The joint behavior of multiple variables. Correlation captures some forms of association but does not fully describe nonlinear relationships or co-movements in extreme events.

[9] **Conditional scenario distribution**: A probability law for future market states over a specified horizon given today's information. Conditioning may require history and latent volatility or regime states, rather than today's prices alone.

$\mathsf{P}_{t,h}$ is a conditional probability distribution over horizon $h$, rather than a valuation function, and must assign its probability to admissible states. Conditioning may be reduced to $X_t$ only when that state contains the required predictive information under an appropriate Markov assumption. Otherwise use history or an augmented state $Z_t$ containing volatility, correlation, and regime probabilities. Today's spot alone generally does not determine a GARCH forecast distribution.

When $Z_t$ is a sufficient predictive state, the conditional model can be expressed as:

$$\mathsf{P}_{t,h}(\cdot \mid \mathcal{F}_t) = \mathsf{P}_{t,h}(\cdot \mid Z_t).$$

Only when $X_t$ itself satisfies this condition may $Z_t = X_t$ be used to obtain the familiar shorthand $P(X_{t+h} \mid X_t)$. For history-conditioned models, $X_{t-k:t}$ denotes the window ending at $t$. Sequence models such as a Transformer can use that window without requiring current prices alone to constitute a sufficient state.

| Model | Principal modeling object | Role in the common framework |
|---|---|---|
| **GARCH** | Conditional return variance and innovation law | Generates marginal future changes given return history and variance state. |
| **DCC-GARCH** | Marginal conditional variances and dynamic correlation | Combines these with a joint innovation law to generate evolving dependence. |
| **Copula** | Joint dependence given marginal distributions | Combines factor-change marginals; temporal dynamics require separate specification. |
| **HMM** | Latent regimes and their transitions | Mixes regime-conditional scenarios using filtered state probabilities. |
| **Transformer** | History windows and other conditioning features | Learns conditional scenarios when equipped with probabilistic outputs or a generation mechanism. |

GARCH commonly models returns rather than price levels. For $R_{t+1} = \log(S_{t+1}/S_t)$, a one-step specification can be written as:

$$\begin{aligned} R_{t+1} &= \mu_{t+1} + \sqrt{v_{t+1}}\, z_{t+1}, \\ v_{t+1} &= \omega + a(R_t - \mu_t)^2 + b v_t, \\ S_{t+1} &= S_t \exp(R_{t+1}). \end{aligned}$$

Here $v$ denotes conditional variance, avoiding confusion with horizon $h$. Specify an innovation law and parameter conditions: for example, $\omega > 0$, $a, b \geq 0$, and independent identically distributed innovations with zero mean and unit variance. In this standard model, $a+b < 1$ is the condition for finite unconditional variance. Conditional return volatility does not automatically equal the implied volatility used in option pricing. Multi-step scenarios generally propagate model states recursively; estimating a conditional variance alone does not define the entire forecast distribution.

A copula representation may apply to factor changes or transformed returns $Y = (Y_1, \ldots, Y_n)$:

$$F_{Y|\mathcal{F}_t}(y_1, \dots, y_n) = C_t\big(F_{1,t}(y_1), \dots, F_{n,t}(y_n)\big)\,.$$

$F_{i,t}$ are conditional marginals and $C_t$ specifies dependence. Generated changes must subsequently reconstruct $X_{t+h} \in \mathcal{X}$. A DCC correlation matrix alone does not determine tail dependence; HMM requires transitions and conditional output laws. A Transformer produces a scenario distribution only with probabilistic prediction or generation machinery; a price point forecast is insufficient.

**Subsequent chapters examine marginal factor dynamics, joint dependence, and scenario generation methods.** These approaches share the valuation and loss-calculation interface. Comparisons should identify the information set, horizon, distributional assumptions, and constraints on state reconstruction.

## From Factor Changes to a P&L Distribution

Let $h > 0$ be the **risk horizon**[10], such as one trading day. When the market state changes from $X_t$ to $X_{t+h}$, applying the valuation function gives the change in portfolio value:

$$\Delta_h V_t = V_{t+h} - V_t = f(t+h, X_{t+h}; \theta) - f(t, X_t; \theta).$$

Throughout this chapter, $\Delta_h$ denotes a finite change from reference time $t$ to $t+h$, rather than a differential. $V_t$ is value and $f$ is the valuation function. Superscripts such as market, carry, and time identify attribution components, not powers. Scenario indices use $(j)$, and $L_{t,h} = -\Delta_h V_t$ is loss over the same horizon. Time inputs in pricing use the model's year-fraction convention; trading-day counts must not be substituted directly into annualized Greeks.

### Why Hold the Valuation Date Fixed?

The **valuation date**[11] is the pricing calculator's "today." It determines how much time an option has left, how long a future cash flow must be discounted, and the date to which bond interest has accrued. The expression $f(t+h, X_{t+h}; \theta)$ changes two inputs at once: market conditions change, and the date inside the calculator advances.

Risk analysis sometimes first asks a more specific question: **If today's portfolio faced a different set of market prices, how much P&L would those market changes produce?** To answer it, leave the calculator's date at today and replace only the scenario prices, rates, exchange rates or volatilities. Each scenario then uses the same valuation time, allowing the market shock's effect to be identified and compared with sensitivities calculated at that fixed date.

For example, suppose today's call has 30 days remaining and the stock price is 100. To examine a fall in the stock price to 95, hold the other market inputs fixed and reprice it as an option valued today with 30 days remaining. Advancing the valuation date to tomorrow would instead leave 29 days and add the effect of one day's time decay. **Holding the date fixed temporarily holds the pricing clock still while market inputs change.**

[10] **Risk horizon**: The future interval over which risk is measured, also called the measurement horizon. A position's P&L distribution generally varies with the horizon and cannot be scaled linearly with time without specifying assumptions.

[11] **Valuation date**: The date the pricing model treats as "today," used to determine remaining maturity, discounting intervals and other time quantities. Data snapshot, job execution and settlement dates require separate definitions. Fixed-date valuation can still use market scenarios spanning a future risk horizon.

Thus, when isolating market-factor movements at valuation date $t$, define $f_t(X;\theta) = f(t, X;\theta)$ and the market-factor P&L:

$$\Delta_h V_t^{\text{market}} = f_t(X_{t+h};\theta) - f_t(X_t;\theta).$$

The fixed input is the function's date argument. The risk horizon $h$ still determines how far ahead market movements are simulated: tomorrow's market state can be valued using today's pricing clock. Holding positions fixed is a separate assumption represented by $\theta$ and should be stated independently.

Whether the date advances follows the measurement purpose. A fixed-date convention isolates market-factor shocks. Estimating the change in portfolio value at a future time advances the date to $t+h$ and requires treatment of the associated time roll and interim cash flows. Fixed-date results are scenario P&L under a specified convention and should be compared with P&L or risk measures using a consistent convention. **The workflow and option case below use the fixed-date convention first, then provide a date-advanced comparison.** This teaching choice does not establish a universal convention for banks or regulatory frameworks; the later discussion also distinguishes internal market-shock analysis from regulatory P&L comparisons.

### Industry Names for the Valuation Date

Trading desks, risk systems and pricing libraries may use different field names. The terms below commonly occur in their respective contexts. Check whether a field controls the pricing clock or identifies a data snapshot.

Table 2: Valuation-date terminology and its context.

| Term | Common name or field | Meaning and context |
|---|---|---|
| Valuation date | **Valuation Date** | The most direct term for the date at which value is calculated. |
| Evaluation date | **Evaluation Date** | Common in pricing libraries; QuantLib's `Settings.evaluationDate`, for example, sets the global evaluation date. |
| Pricing date | **Pricing Date** | Used in pricing workflows; contractual documents may also use it for the date an issue price or terms are determined. |
| As-of date | **As-of Date** | Common for market-data, position and reporting snapshots; its use as the valuation date depends on the system definition. |
| Valuation time or timestamp | **Valuation Time / Valuation Timestamp** | A more precise intraday valuation point, requiring a time zone and market-data cutoff. |

A risk platform may also use **Risk Date / Risk As-of Date** for its reporting reference date; its relationship to valuation, position and market-data dates must be specified. **Run Date / Calculation Date** may identify when a job executes: running yesterday's end-of-day risk report today gives different dates. **Trade Date** records execution, while **Settlement Date**, and commonly **Value Date** in FX, identify settlement or the date funds become effective. Each serves its own purpose. In the formulas above, $t$ is the date that controls the pricing clock.

For realized future P&L, the treatment of interim cash flows, position changes, and transaction costs must also be specified. For example, a coupon payment means that ending value minus starting value alone does not capture the full return on a bond. In this introductory discussion, positions are fixed and interim cash flows and transaction costs are excluded, so $\Delta_h V_t$ represents **profit and loss (P&L)**. Positive values indicate gains and negative values indicate losses. The corresponding **loss variable** is:

$$L_{t,h} = -\Delta_h V_t.$$

Thus, $L_{t,h} > 0$ indicates a loss. The variable may also be negative, indicating a gain. Either P&L or loss may be used as the model output, provided the sign convention is explicit.

## Market Risk P&L Decomposition

Freezing or advancing the date is a valuation setting. A bank's **P&L Attribution Framework**[12] must additionally explain the sources of desk P&L. Under the fixed-position, no-interim-cash-flow simplification above, an attribution can be written as:

$$\Delta_h V_t = \Delta_h V_t^{\text{market}} + \Delta_h V_t^{\text{carry}} + \Delta_h V_t^{\text{time}} + \epsilon_{t,h}.$$

The market component uses the preceding fixed-date shock definition. The remaining time roll is allocated between carry and pure time effects under explicit conventions. Carry may include accrual or roll along specified curves already reflected in valuation; option time effects are commonly explained locally by Theta. The residual $\epsilon_{t,h}$ reconciles the attribution to total value change and may reflect approximation error, unexplained items, or data issues. An exact, complete revaluation decomposition can have zero residual; it must not be treated as negligible noise by default.

**Attribution components must not overlap.** Some institutions include Theta within a broad definition of carry; in that case the same effect cannot also be added separately. Rate shocks and curve roll require consistent curve coordinates and roll rules. Applying a market shock before rolling time can allocate interaction effects differently from the reverse ordering, but total P&L must remain unchanged.

For an actual desk with trades and cash flows, value change alone does not represent complete P&L. A schematic ledger decomposition is:

$$\begin{aligned}\text{P\&L}_{t,h}^{\text{desk}} = \text{P\&L}_{t,h}^{\text{market}} + \text{P\&L}_{t,h}^{\text{carry}} + \text{P\&L}_{t,h}^{\text{time}} \\ + \text{P\&L}_{t,h}^{\text{trade}} + \text{P\&L}_{t,h}^{\text{fees}} + \text{P\&L}_{t,h}^{\text{other}}.\end{aligned}$$

The trade component requires reconciliation of new, modified, or closed positions and their cash flows. Fees are signed P&L, with expenses generally negative. Coupons and expiry payments enter the specified cash-flow categories without double counting accrual or mark changes. This expression classifies attribution sources; it does not treat every cash flow as a new loss or establish a universal accounting identity without ledger reconciliation.

The framework links three purposes: **VaR/ES** construct future loss distributions for a specified P&L convention; **P&L Explain** identifies sources of realized P&L; **FRTB PLA** compares two

[12] **P&L attribution framework**: A reconciliation of total P&L to sources under explicit position, market, time, trade, and cash-flow conventions. Component boundaries and interaction allocations must be stated; daily P&L Explain and regulatory PLA compare different objects.

P&L series under regulatory definitions. The following distinctions establish the connection to the later discussion:

| Series | Meaning | Relationship to attribution |
|---|---|---|
| **APL**: Actual P&L | Desk actual P&L under applicable rules | Reconciles actual trades and relevant sources; it is not automatically the sum of every ledger item. |
| **HPL**: Hypothetical P&L | Revaluation of previous closing positions held constant | Isolates intraday trading effects; static positions do not automatically freeze the date. |
| **RTPL**: Risk-Theoretical P&L | P&L generated using risk-model factors and valuation | Compared with HPL to assess whether factors and valuation adequately explain portfolio P&L. |

Daily Greek-based explanation differs from regulatory PLA, and backtesting must also use the P&L series required by the applicable rules. The time-decay section develops the treatment of time in APL, HPL, and RTPL and their regulatory comparisons.

Core Modeling Objective

A market risk model describes the **probability distribution** of future P&L, rather than only a price forecast or an average P&L. Given the information $\mathcal{F}_t$ available at time $t$, this objective can be expressed through a conditional distribution function:

$$F_{\Delta_h V_t|\mathcal{F}_t}(z) = \Pr(\Delta_h V_t \le z \mid \mathcal{F}_t)\,.$$

Here, $z$ is an arbitrary P&L threshold. The notation $P(\Delta_h V_t)$ is sometimes used as shorthand for this distribution; it should not be read as the probability that a continuous variable takes one exact value.

The basic chain is: **future joint distribution of risk factors → scenario valuation → P&L or loss distribution → risk measures**. The following sections develop the complete workflow and use Value at Risk (VaR) and Expected Shortfall (ES) to summarize an option loss distribution's tail. The distribution here describes possible real-world P&L outcomes; its purpose differs from that of a risk-neutral probability distribution used in derivative pricing. See McNeil, Frey, and Embrechts (2015), and Jorion (2007).

### Example 0.1: A Single Stock Position

Suppose an institution holds $q = 1{,}000$ shares at a current price of $S_t = 100$ CNY. This example indexes time by trading days and sets $h = 1$. Ignoring dividends and transaction costs, the valuation function is $V_t = qS_t$, and the position is worth 100,000 CNY. To illustrate the modeling logic, assume only the following two scenarios for the next trading day. The probabilities are teaching assumptions, rather than data-based forecasts.

| Scenario | Probability | $S_{t+1}$ (CNY) | $\Delta_h V_t$ (CNY) | $L$ (CNY) |
|---|---|---|---|---|
| Up | 0.60 | 102 | +2,000 | −2,000 |
| Down | 0.40 | 97 | −3,000 | +3,000 |

The valuation function gives $\Delta_h V_t = q(S_{t+1} - S_t)$, so:

$$\mathbb{E}[\Delta_h V_t \mid \mathcal{F}_t] = 0.60 \times 2{,}000 + 0.40 \times (-3{,}000) = 0.$$

Although expected P&L is zero, the portfolio still has a 40% probability of losing 3,000 CNY. **Average P&L cannot substitute for the risk distribution.** If the position is short, with $q = -1{,}000$, the same upward price scenario produces a loss of 2,000 CNY, illustrating why an adverse movement depends on the direction of the position.

This section represents market risk as potential loss arising when market-state changes pass through a valuation function. A model must specify its risk factors, positions and valuation rules, risk horizon, and P&L convention before the distribution of future P&L can be interpreted meaningfully.

## 0.1.2 From Risk Factor Changes to Portfolio Loss Distribution

The abstraction $V = f(X)$ becomes operational only after the institution specifies what is held, which market variables are modeled, how scenarios are valued, and what constitutes a loss. This section makes those choices explicit. Throughout, positions are held constant, all values use a common reporting currency, and there are no interim cash flows or transaction costs unless stated otherwise.

### Step 1: Portfolio Value as a Function of Risk Factors

A financial product's value is conditional on a market state. For a portfolio of $m$ instruments, write:

$$V_t = f(t, X_t; \theta) = \sum_{a=1}^{m} q_a v_a(t, X_t; \theta_a), \qquad X_t = \begin{bmatrix} x_{1,t} \\ x_{2,t} \\ \vdots \\ x_{n,t} \end{bmatrix} \in \mathcal{X}.$$

Here $q_a$ is a signed position quantity, $v_a$ is the value per unit in the reporting currency, and $\theta_a$ contains contractual terms and other fixed specifications. Each $x_{i,t}$ represents a market variable, such as an **equity spot price**, an **interest rate curve** node, an **FX rate**, a **credit spread**, or a point on a **volatility surface**[13]. Curves and surfaces require multiple coordinates or a specified parameterization; the chosen factor representation must permit internally consistent reconstruction for pricing.

Factor mapping is an economic modeling decision. Omitting implied volatility from the state of an option portfolio prevents the model from representing losses caused by changes in option market quotes. Including it without defining the expiry, strike or moneyness convention leaves the meaning of a shock ambiguous. Similarly, foreign-currency valuations must be converted using scenario exchange rates rather than today's exchange rate when FX is a modeled exposure.

[13] **Volatility surface**: Implied volatility indexed by expiry and strike or moneyness. A scenario must specify how the surface is moved and reconstructed; a flat-volatility shock is a teaching simplification.

### Step 2: Market Evolution and Portfolio Revaluation

Over a horizon $h$, the uncertain market evolution $X_t \to X_{t+h}$ leads to uncertain scenario values. **The risk horizon specifies the interval over which markets change; it does not automatically advance the pricing calculator's date.** This chapter uses a fixed-date market-shock convention as its baseline:

$$f_t(X;\theta) = f(t,X;\theta), \qquad \Delta_h V_t^{\text{market}} = f_t(X_{t+h};\theta) - f_t(X_t;\theta).$$

Each scenario substitutes new market inputs while holding the valuation clock at $t$. This is an explicit measurement convention: compare the value of today's portfolio across market states. The engine may simulate future factors by Monte Carlo or apply historical factor changes to today's state. Choosing Monte Carlo or historical simulation does not itself determine the treatment of time decay.

These scenarios follow the chosen conditional model $\mathsf{P}_{t,h}(\cdot \mid \mathcal{F}_t)$, with each state belonging to $\mathcal{X}$. Replacing GARCH, a copula, or another scenario model while retaining valuation and P&L conventions isolates how distributional assumptions affect risk measures. A state that cannot be reconstructed validly has not yet completed the conversion from a probability model to pricing inputs.

If the objective instead becomes the value change from holding the portfolio until $t+h$, advance the valuation date as well:

$$\Delta_h V_t = V_{t+h} - V_t = f(t+h, X_{t+h};\theta) - f(t, X_t;\theta).$$

This definition reflects both market movements and time passage. The distinction matters for finite-lived instruments, accrued interest and scheduled payments. These are different P&L definitions; holding positions constant and freezing the valuation date are also separate assumptions. The no-interim-cash-flow simplification remains in force here. Coupons or expiry payments during the interval must enter holding-period P&L under a consistent valuation convention. An exact bridge between the two value changes is:

$$\begin{aligned} \Delta_h V_t^{\text{roll}} &= f(t+h, X_{t+h};\theta) - f(t, X_{t+h};\theta), \\ \Delta_h V_t &= \Delta_h V_t^{\text{market}} + \Delta_h V_t^{\text{roll}}. \end{aligned}$$

The roll component is the complete value change from advancing the clock at the future market state. The identity applies the market shock before rolling time and provides the exact total to allocate between carry and pure time effects in the preceding framework. It does not label the entire roll as Theta. A different ordering can allocate interactions differently while leaving total value change unchanged.

The Quantitative Question

**How uncertain market movements translate into uncertain portfolio value changes.**
The task requires a joint model for future factors and a valuation map for the portfolio. An accurate pricing model alone does not specify the probability of tomorrow's losses; a factor forecast alone does not specify how much a nonlinear portfolio loses.

### Step 3: From P&L to Loss Distribution

P&L must first be assigned a convention. For the chapter's baseline market-factor P&L, risk reporting converts it into loss:

$$\mathrm{P\&L}_{t,h}^{\text{market}} = \Delta_h V_t^{\text{market}}, \qquad L_{t,h}^{\text{market}} = -\Delta_h V_t^{\text{market}}.$$

For the date-advanced holding-period convention, the corresponding definition is $L_{t,h} = -\Delta_h V_t$. The sign convention makes adverse outcomes positive and places severe losses in the upper tail. A negative loss remains a gain; it does not discard profitable scenarios. Below, $L_j$ abbreviates scenario loss under the baseline convention.

Suppose the scenario engine produces $N$ joint states $X_{t+h}^{(j)}$. Each passes through the same valuation and loss rules:

$$V_{\text{frozen}}^{(j)} = f_t(X_{t+h}^{(j)}; \theta), \qquad L_j = V_t - V_{\text{frozen}}^{(j)}, \quad j = 1, \dots, N.$$

The outcomes $L_1, L_2, \dots, L_N$, together with their probabilities, form a **loss distribution**. Equal-weight scenarios imply the empirical distribution:

$$\widehat{F}_L(\ell) = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\{L_j \le \ell\}.$$

For confidence $\alpha \in (0, 1)$, conditional on today's information, define:

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell : F_L(\ell) \ge \alpha\}, \qquad \mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \mathrm{VaR}_u(L)\, du.$$

**Value at Risk**[14] identifies a loss quantile. **Expected Shortfall**[15] averages the upper tail and therefore describes the severity beyond that quantile. For a continuous distribution with suitable integrability, ES can be expressed as the conditional mean above VaR; the quantile-integral definition also handles probability mass at the boundary. With discrete or weighted scenarios, tail averaging must allocate the correct probability at that boundary.

Neither quantity is a maximum possible loss. Both depend on the horizon, positions, factor model, valuation rules, and P&L convention. A deterministic stress scenario gives a scenario loss; it does not by itself provide probabilities sufficient to calculate VaR or ES.

The Market Risk Quant Workflow

**Risk Factors → Pricing → Revaluation → P&L → Risk Metrics**
Specify exposures and joint scenarios; establish today's valuation; revalue under each scenario; compute signed P&L and loss; aggregate the distribution into VaR, ES, and complementary stress results. Every reported risk number should be traceable through this chain.

[14] **Value at Risk (VaR)**: A quantile of the loss distribution at a specified confidence and horizon. It is a threshold, not the largest possible loss or the mean loss beyond the threshold.

[15] **Expected Shortfall (ES)**: The average of loss quantiles over the upper $1 - \alpha$ probability tail. For discrete samples, correct boundary weighting avoids averaging the wrong tail probability.

## 0.1.3 Case Study: From Option Pricing to Market Risk Measurement

Consider a long European call[16] on a non-dividend-paying equity. It illustrates why the risk pipeline must include a pricing function: a change in the underlying price does not translate one-for-one into a change in option value.

### Step 1: Option Pricing Function

In reduced notation, the option value is:

$$V = f(S, \sigma, r, T).$$

Here $T$ denotes **remaining** time to maturity. To avoid confusing it with an expiry date, the formulas below use $\tau$ for remaining maturity and $t_{\text{exp}}$ for the fixed expiry, so $\tau = t_{\text{exp}} - t$.

| Input | Meaning | Role in this example |
|---|---|---|
| $S$ | Underlying spot price | Stochastic market risk factor |
| $\sigma$ | Annualized implied volatility | Stochastic market risk factor |
| $r$ | Continuously compounded interest rate | Stochastic market risk factor |
| $T$ or $\tau$ | Time to maturity, in years | Deterministic time state for a fixed expiry |

Maturity is a valuation input but is not a randomly shocked market factor in this example. Strike $K$ and the position quantity are fixed contractual and portfolio inputs. In production, one rate and one volatility usually become discount/forward curves and a volatility surface.

For transparent calculations, use the Black-Scholes model:

$$C(S, \sigma, r, \tau; K) = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),$$
$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

$\Phi$ is the standard normal distribution function. This model uses risk-neutral valuation[17] under its pricing assumptions; the future market scenarios used for risk measurement are modeled under real-world probabilities. Using the same symbol $\sigma$ does not make the two probability measures interchangeable. The formula and its assumptions originate in Black and Scholes (1973).

Fix $K = 100$, $r_t = 2\%$, and $\tau_t = 30/365$. Use ACT/365, no dividends, no funding or transaction costs, and no interim payments. All values are per unit of underlying, in currency units, without a contract multiplier. A position covering $Q$ units multiplies each valuation and each scenario P&L by the signed quantity $Q$; positive scaling also scales VaR and ES.

[16] **European call**: A contract giving its holder the right to purchase the underlying at strike $K$ on the expiry date, with payoff $\max(S_{t_{\text{exp}}} - K, 0)$ per underlying unit.

[17] **Pricing and real-world measures**: Risk-neutral probabilities support arbitrage-consistent derivative pricing under the model. Real-world probabilities describe the future market outcomes relevant to loss forecasting. They serve different purposes.

The following functions implement the pricing formula for $S, K, \sigma > 0$ and $\tau \geq 0$, with decimal rates and volatilities and maturity in years. At expiry they return intrinsic value. Python accepts scalars or NumPy arrays; C++ prices one instrument under one scenario.

**Python**

```python
import numpy as np
from scipy.special import ndtr


def call_price(S, K, sigma, r, tau):
    if tau == 0:
        return np.maximum(S - K, 0.0)
    d1 = (np.log(S / K) + (r + sigma**2 / 2) * tau)
    d1 /= sigma * np.sqrt(tau)
    d2 = d1 - sigma * np.sqrt(tau)
    return S * ndtr(d1) - K * np.exp(-r * tau) *
    ↪  ndtr(d2)
```

**C++**

```cpp
#include <cmath>

inline double call_price(double S, double K,
↪  double sigma,
                         double r, double tau) {
    if (tau == 0.0) return S > K ? S - K : 0.0;
    const auto Phi = [](double x) {
        return 0.5 * std::erfc(-x /
        ↪  std::sqrt(2.0));
    };
    const double root = std::sqrt(tau);
    const double d1 = (std::log(S / K)
        + (r + sigma * sigma / 2) * tau) / (sigma
        ↪  * root);
    return S * Phi(d1)
        - K * std::exp(-r * tau) * Phi(d1 - sigma
        ↪  * root);
}
```

## Step 2: Risk Factor Shock and Revaluation

Use the following joint market shock over one calendar day. First freeze the valuation date, then advance it as a comparison; both columns use identical market inputs:

| Input | Today | Date frozen | Date advanced |
|---|---|---|---|
| Spot price $S$ | 100 | 95 | 95 |
| Implied volatility $\sigma$ | 20% | 25% | 25% |
| Interest rate $r$ | 2% | 2% | 2% |
| Remaining maturity | 30/365 | 30/365 | 29/365 |

At the fixed date, invoke the full pricing model again:

$$V_{\text{frozen}} = f(S_{t+1}, \sigma_{t+1}, r_{t+1}, \tau_t).$$

Applying the pricing function yields:

$$V_t = 2.368335, \qquad V_{\text{frozen}} = 1.009287,$$
$$\Delta_h V_t^{\text{market}} = -1.359048, \qquad L^{\text{market}} = 1.359048.$$

The long call loses value despite higher implied volatility: the adverse spot movement dominates the beneficial volatility effect. This is already full revaluation, with no advancement of the option's pricing clock.

To calculate the holding-period value separately, use:

$$V_{t+1} = f(S_{t+1}, \sigma_{t+1}, r_{t+1}, \tau_t - 1/365) = 0.971385,$$
$$\Delta_h V_t = -1.396950, \qquad L = 1.396950.$$

Their difference, $-0.037902$, is the exact one-day roll effect **at the shocked market state**. It need not equal today's Theta multiplied by one day. This comparison demonstrates that **full revaluation specifies how prices are calculated; the P&L convention determines whether time advances.**

In code, this distinction changes only the remaining-maturity argument; loss is initial value less scenario value in both cases:

**Python**

```
K, r, tau, h = 100.0, 0.02, 30 / 365, 1 / 365
v0 = call_price(100, K, 0.20, r, tau)
v_frozen = call_price(95, K, 0.25, r, tau)
v_rolled = call_price(95, K, 0.25, r, tau - h)
stress_loss_frozen = v0 - v_frozen
stress_loss_rolled = v0 - v_rolled
```

### Step 3: Explain Sources of Option P&L

Greeks[18] provide a local interpretation of revaluation. Define today's sensitivities:

$$\delta = \frac{\partial C}{\partial S}, \quad \Gamma = \frac{\partial^2 C}{\partial S^2}, \quad \nu = \frac{\partial C}{\partial \sigma}, \quad \rho = \frac{\partial C}{\partial r}, \quad \Theta = \frac{\partial C}{\partial t} = -\frac{\partial C}{\partial \tau}.$$

At the fixed date, retain spot curvature and first-order volatility and rate effects to approximate market-factor P&L:

$$\Delta_h V_t^{\text{market}} \approx \underbrace{\delta\,\Delta S}_{\text{Delta}} + \underbrace{\tfrac{1}{2}\Gamma(\Delta S)^2}_{\text{Gamma}} + \underbrace{\nu\,\Delta\sigma}_{\text{Vega}} + \underbrace{\rho\,\Delta r}_{\text{Rho}}.$$

The date-advanced approximation additionally includes a time term:

$$\Delta_h V_t \approx \delta\,\Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2 + \nu\,\Delta\sigma + \rho\,\Delta r + \Theta h.$$

The Delta term is **Delta sensitivity multiplied by the spot change**, not the spot change alone. Here $h = 1/365$ is elapsed calendar time; $\Delta\tau = -h$ when the date advances, but $\Delta\tau = 0$ when it is frozen. All Greeks are evaluated at today's state. Volatility and rates use decimals: $20\% \to 25\%$ means $\Delta\sigma = 0.05$. Vega per one percentage point is $\nu/100$; annual Theta must be multiplied by a year fraction.

| Source | Initial sensitivity | Applied change | Approximate P&L |
|---|---:|---:|---:|
| Delta | 0.522862 | $\Delta S = -5$ | $-2.614311$ |
| Gamma | 0.069463 | $(\Delta S)^2 = 25$ | $+0.868284$ |
| Vega | 11.418534 | $\Delta\sigma = 0.05$ | $+0.570927$ |
| Theta (date advanced only) | $-14.890907$ per year | $h = 1/365$ | $-0.040797$ |
| Rho | 4.102839 | $\Delta r = 0$ | 0 |

[18] **Greeks**: Local derivatives of an option valuation function. Their interpretation requires the input convention, units and sign of the position. They change as the market state and remaining maturity change.

| P&L convention | Approximation | Full revaluation | Residual |
|---|---|---|---|
| Date frozen, no Theta | −1.175099 | −1.359048 | −0.183948 |
| Date advanced, today's Theta added | −1.215896 | −1.396950 | −0.181054 |

**Delta** measures the first-order exposure to the underlying. **Gamma** captures local curvature; the long call's positive Gamma offsets part of the loss predicted by linear Delta alone. **Vega** measures sensitivity to implied volatility; the volatility increase benefits this long call. **Theta** measures the calendar-time effect at unchanged inputs. **Rho** measures rate sensitivity and contributes zero because the scenario leaves rates unchanged.

The residual is full-revaluation P&L less the displayed approximation. A five-unit spot fall and five-percentage-point volatility increase are substantial for a short-dated option. Higher-order terms, volatility curvature and spot-volatility cross effects can therefore matter. Vanna and Volga[19] are examples of omitted terms. The residual is not automatically a pricing error, nor does this truncated Taylor expansion uniquely allocate all interactions among factors.

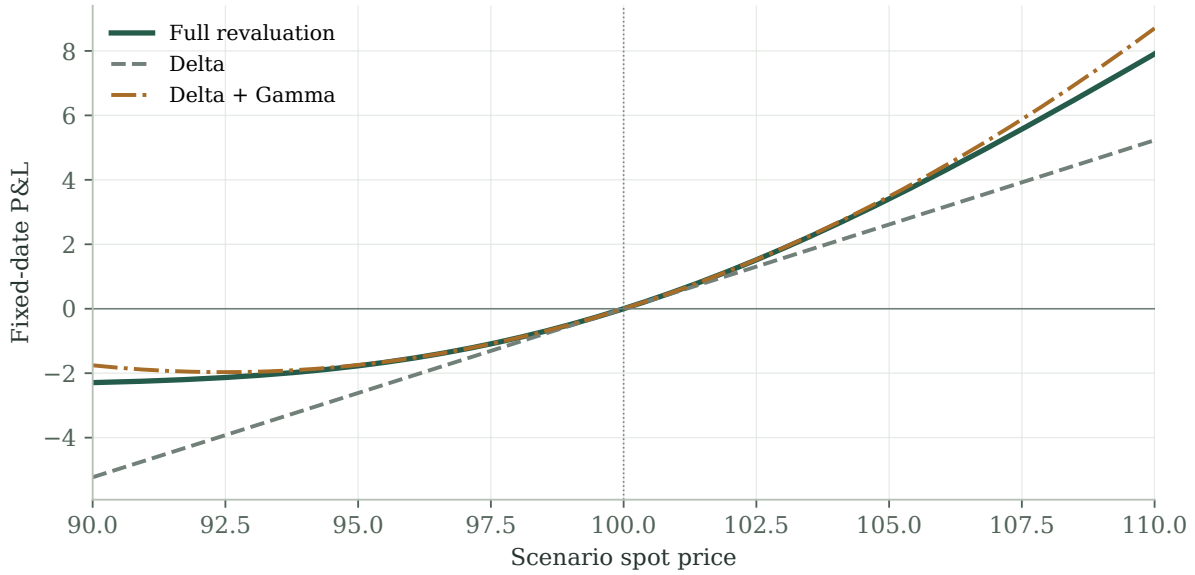



Figure 1: Option P&L for spot shocks alone. The valuation date, implied volatility (20%) and interest rate (2%) are fixed; all sensitivities are evaluated at the initial spot of 100.

The curves agree near the initial state and separate as shocks grow. Delta supplies a tangent and Gamma adds local curvature; both remain local approximations. This figure isolates spot shocks, unlike the earlier stress scenario that changes spot and implied volatility together.

## Step 4: Generate Scenarios and Measure Portfolio Risk

To move from one stress result to a loss distribution, generate $N = 100{,}000$ equally weighted joint scenarios. The following assumptions are deliberately pedagogical, not calibrated forecasts. Let $Z_1, Z_2, Z_3$ be independent standard normals and set:

[19] **Vanna and Volga**: Vanna is $\partial^2 C/(\partial S \partial \sigma)$; Volga is $\partial^2 C/\partial \sigma^2$. Their Taylor contributions are respectively Vanna $\Delta S \Delta \sigma$ and $\frac{1}{2}$Volga$(\Delta \sigma)^2$.

$$S^{(j)} = 100 \exp\left[-\tfrac{1}{2}(0.0125)^2 + 0.0125 Z_1^{(j)}\right],$$
$$\sigma^{(j)} = 0.20 \exp\left[0.05(-0.6 Z_1^{(j)} + 0.8 Z_2^{(j)})\right], \qquad r^{(j)} = 0.02 + 0.0001 Z_3^{(j)}.$$

Spot log returns have a standard deviation of 1.25%; log changes in implied volatility have a standard deviation of 5% and a correlation of $-0.6$ with spot log returns. This is a relative volatility shock, distinct from a five-percentage-point change in implied volatility. Rate changes have a standard deviation of one basis point. Exponentiation keeps spot and implied volatility positive. Rate shocks are independent of the other shocks in this illustration[20].

The correlation matrix below orders equity log returns, log changes in implied volatility and absolute rate changes. Cholesky decomposition transforms independent draws into correlated shocks. Each column is one joint scenario, subsequently repriced with either a frozen or an advanced date.

**Python**

```
rng = np.random.default_rng(20261010)
R = np.array([[1.0, -0.6, 0.0],
              [-0.6, 1.0, 0.0],
              [0.0, 0.0, 1.0]])
eps = np.linalg.cholesky(R) @ rng.standard_normal((3, 100_000))
S = 100 * np.exp(-0.5 * 0.0125**2 + 0.0125 * eps[0])
sigma = 0.20 * np.exp(0.05 * eps[1])
rates = 0.02 + 0.0001 * eps[2]
loss_frozen = v0 - call_price(S, K, sigma, rates, tau)
loss_rolled = v0 - call_price(S, K, sigma, rates, tau - h)
```

The left point cloud slopes downward, whereas the right has no corresponding directional structure. Dependence determines which market changes tend to occur together and thus how Delta, Vega and Rho exposures combine at portfolio level.

Under the baseline convention, retain remaining maturity 30/365 in every scenario and calculate market-shock loss:

$$L_j = 2.3683354166 - C(S^{(j)}, \sigma^{(j)}, r^{(j)}, 30/365; 100).$$

With NumPy's generator seed 20261010, inverse-CDF empirical VaR and quantile-integral ES give:

[20] **Joint shock modeling**: Independent rate shocks are a teaching simplification. This example already links equity and volatility through the shared $Z_1$; dependence involving rates also matters in practice.

Estimate a correlation matrix $R$ from date-aligned factor changes over the same risk horizon, then transform independent standard normal draws into correlated shocks:

$$Z \sim \mathcal{N}(0, I),$$
$$\varepsilon = BZ,$$
$$BB^{\mathsf{T}} = R.$$

Then $\mathrm{Cov}(\varepsilon) = R$; Cholesky decomposition gives $B$ for positive definite $R$. Scale components by their shock standard deviations, add means, and reconstruct market states. Nonzero correlations involving rates in $R$ introduce rate dependence. Check that $R$ is a valid positive semidefinite correlation matrix; it describes correlations of **changes**.

Historical simulation can resample whole vectors of changes from the same date, retaining sample dependence. Multivariate Student-$t$ distributions or $t$-copulas can represent tail dependence of simultaneous extremes in a crisis, supplemented by stress scenarios. Volatilities and dependence can also be updated over time. Dependence affects hedging effectiveness and the tail of portfolio losses.

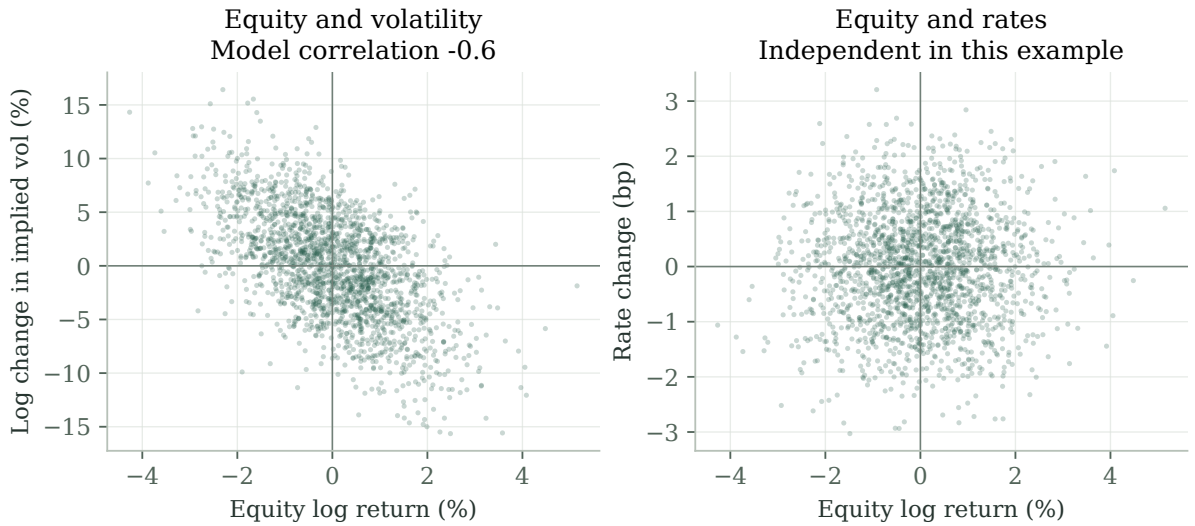



Figure 2: Factor changes from the same simulated scenarios. The left panel shows negative equity-volatility correlation; the right shows the independent rate shocks assumed here. Each panel displays 2,000 scenarios.

| Valuation and P&L convention | $\text{VaR}_{97.5\%}$ | $\text{ES}_{97.5\%}$ |
|---|---|---|
| Full revaluation, date frozen | 0.954271 | 1.087272 |
| Delta-Gamma + Vega + Rho, date frozen | 0.944457 | 1.071780 |
| Full revaluation, date advanced | 0.993001 | 1.124850 |
| Same approximation, plus today's Theta | 0.985254 | 1.112577 |

The first two rows compare valuation methods at the fixed date. The last two reduce maturity to 29/365 or add today's Theta as holding-period comparisons. All rows use identical simulated market states, so their differences isolate valuation approximation and the treatment of time within this experiment. Adding today's constant Theta shifts approximate losses by 0.040797 and shifts both VaR and ES by that amount. Exact date advancement changes each scenario by its own roll effect and need not produce a constant shift.

For these 100,000 equal-weight outcomes, the 97.5% inverse-CDF VaR is the 97,500th loss in ascending order. ES averages the largest 2,500 losses. This connects the reported numbers directly to the scenario loss sample; other confidence levels may require fractional boundary weights.

The following implementations assume a nonempty, finite, equally weighted loss sample and $0 < \alpha < 1$. Tail weights follow probability mass and therefore also handle a non-integer number of tail observations.

**Python**

```python
def empirical_risk(losses, alpha=0.975):
    x = np.sort(np.asarray(losses, dtype=float))
    var = np.quantile(x, alpha,
    ↪  method="inverted_cdf")
    mass = len(x) * (1 - alpha)
    weights = np.clip(mass - np.arange(len(x)),
    ↪  0.0, 1.0)
    es = np.dot(x[::-1], weights) / weights.sum()
    return float(var), float(es)


var_frozen, es_frozen = empirical_risk(loss_frozen)
var_rolled, es_rolled = empirical_risk(loss_rolled)
```

**C++**

```cpp
#include <algorithm>
#include <cmath>
#include <utility>
#include <vector>

inline std::pair<double, double> empirical_risk(
    std::vector<double> x, double alpha = 0.975) {
    std::sort(x.begin(), x.end());
    const double mass = x.size() * (1 - alpha);
    double sum = 0.0, weight = 0.0;
    for (std::size_t i = 0; i < x.size(); ++i) {
        const double w = std::clamp(mass -
        ↪  double(i), 0.0, 1.0);
        sum += w * x[x.size() - 1 - i];
        weight += w;
    }
    const auto k = std::size_t(std::ceil(alpha *
    ↪  x.size()));
    return {x[k - 1], sum / weight};
}
```

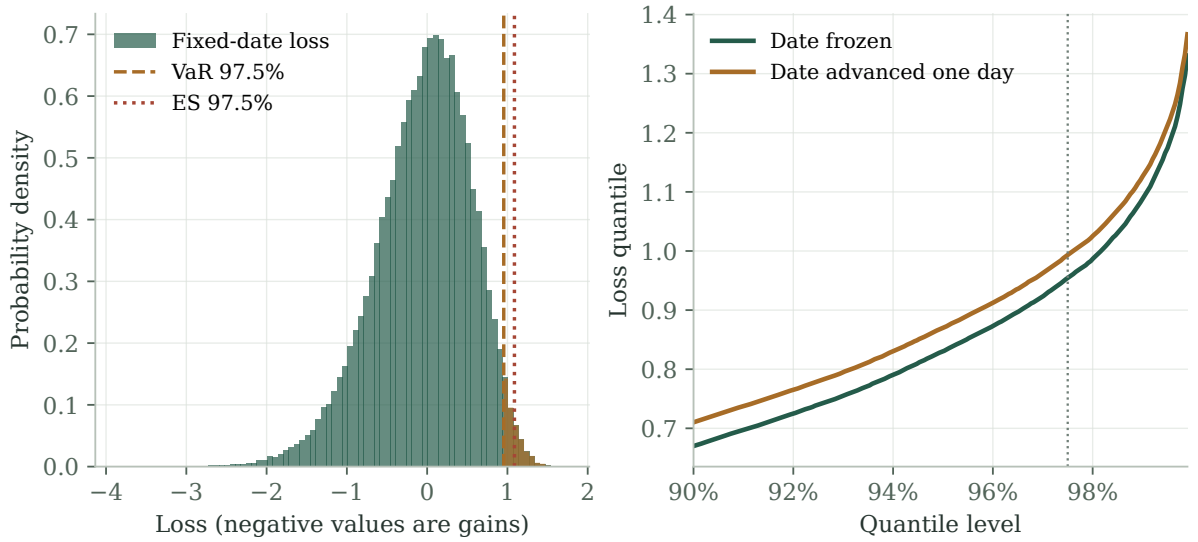



Figure 3: Loss distribution and time convention. The left panel shows fixed-date loss density, with the upper 2.5% tail shaded in gold and VaR and ES marked separately. The right compares loss quantiles under the two date conventions using identical market scenarios.

VaR marks the start of the tail and ES summarizes its mean loss; the ES line is not a second quantile threshold. The right panel shows that advancing the date changes the loss distribution. In this example, decay increases the long call's tail losses, with a roll effect that varies by scenario.

The long call has a nonnegative value, so its exact loss cannot exceed its initial premium in this no-cash-flow example. A local polynomial approximation need not preserve that bound outside its useful shock range. More scenarios reduce sampling error but do not correct omitted factors, inappropriate pricing assumptions, or unrealistic tail dependence.

An institutional implementation additionally reconciles today's value to the approved pricing source, checks scenario coverage and curve/surface construction, records valuation failures, and investigates unexplained P&L. The reported result must identify its confidence level, horizon, portfolio snapshot, probability model, and time convention. Subsequent backtesting must compare the estimate with a consistently defined observed P&L series.

## 0.1.4 Why Time Decay Matters in Option Risk

An option grants a right that expires. Even if spot, implied volatility and rates remain unchanged, the time available for future favorable movements changes. For a fixed expiry:

$$\tau \longrightarrow \tau - h, \qquad \Delta_h V_t^{\text{time}} = C(S, \sigma, r, \tau - h) - C(S, \sigma, r, \tau) \approx \Theta h.$$

This is **time decay**[21]. The market state can remain constant while the calendar-time argument of the valuation function changes. For the non-dividend-paying long call with positive rates used here, Theta is negative before expiry. The statement does not imply that every option or every option position has negative Theta; short positions reverse the sign, and other contracts or market conditions require their own analysis.

### An Illustrative One-Day Loss

Suppose a separate illustrative option has the following values, with all market inputs unchanged:

| Date | Option value | Market inputs |
|---|---|---|
| Day 0 | 2.50 | Fixed |
| Day 1 | 2.45 | Fixed |

Then $\Delta_h V_t = 2.45 - 2.50 = -0.05$ and $L = 0.05$. The loss comes from the change in remaining maturity and is explained locally by Theta. These premiums illustrate the mechanism; they are not the prices calculated from the preceding call's parameters.

For that preceding call, exact unchanged-market repricing instead gives:

$$C(100, 0.20, 0.02, 30/365) = 2.368335,$$
$$C(100, 0.20, 0.02, 29/365) = 2.327215, \qquad L^{\text{time}} = 0.041121.$$

Today's Theta predicts a loss of 0.040797. The small difference reflects the change in Theta during the finite interval; a derivative is a local approximation, not an exact finite-day price change.

The solid line declines even without a market shock. The dashed line is tangent at the initial state and diverges as time passes, demonstrating why today's constant Theta cannot replace exact repricing over an entire holding period.

[21] **Time decay**: Value change caused by passage of time with market inputs held fixed. Theta approximates it locally; a finite-period roll is obtained by repricing at the later date.

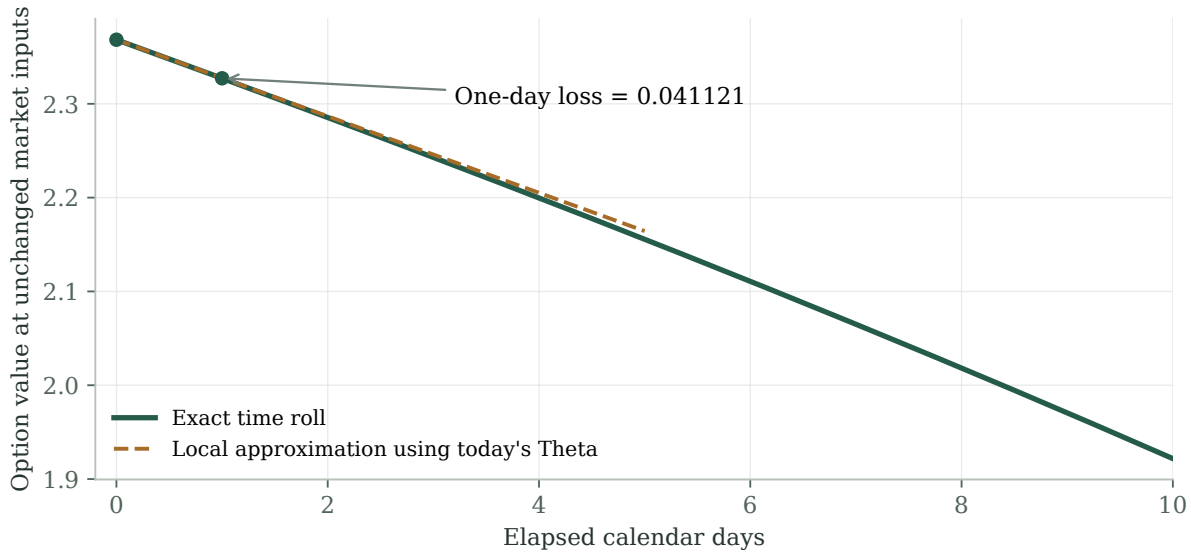



Figure 4: Option value as time passes with market inputs fixed. The solid line is exact repricing and the dashed line uses the initial Theta locally. Markers identify values at day 0 and day 1.

## Time Decay, Risk and the P&L Convention

At today's unchanged market state, time passage is deterministic. Nevertheless, future option losses combine this roll with random market movements, and the roll effect itself varies across shocked states. Near expiry, the sensitivity of value to spot and volatility can also change rapidly. Weekend horizons and expiry crossings therefore require explicit calendar and payoff treatment rather than a mechanical one-day Theta adjustment.

If a risk statistic describes future marked portfolio losses at $t + h$, time passage belongs in that valuation, along with any relevant intervening cash flows. If the statistic describes losses due only to market-factor shocks at today's date, time passage is excluded by construction and can be reported separately. For actual or hypothetical P&L comparisons, institutions must document and reconcile the chosen convention; a fixed-date shock estimate and an observed P&L including decay need not be directly comparable.

**Time effects remain important sources of economic portfolio P&L.** Under the preceding attribution framework, daily P&L Explain reconciles market, carry, pure time effects, and other sources without overlapping categories. A fixed-date market-shock measure separates pure time effects, while risk management still identifies and monitors them. The local time P&L is $\Theta h$ and the corresponding loss is $-\Theta h$; units and horizon must be stated.

## From Internal Measurement to Regulatory P&L Comparisons

The APL, HPL, and RTPL series introduced above[22] must follow their specific definitions in regulatory comparisons. Basel MAR32.25 defines HPL by revaluing the previous day's closing positions with the current day's market data, excluding intraday trading and new or modified

[22] **APL, HPL and RTPL**: Regulatory actual P&L, hypothetical P&L on the previous closing positions, and risk-theoretical P&L generated with the risk model's factors and valuation engines. Included items follow the applicable rules; "hypothetical" does not automatically mean frozen time. The PLA test compares HPL with RTPL, unlike daily Greek-based attribution.

trades. **Static positions do not imply a static valuation date.** MAR32.27 requires time-passage P&L to be included in APL and treated consistently in HPL and RTPL; its footnote explicitly lists Theta, carry and funding costs. See Basel Committee on Banking Supervision, MAR32, paragraphs 32.25–32.27.

Backtesting under MAR32.4 compares one-day VaR with APL and HPL. The regulatory P&L Attribution (PLA) test under MAR32.24 compares RTPL with HPL. These serve different purposes from daily Greek-based P&L explanation. These requirements mean that a time convention cannot be inferred from the label "bank VaR," nor can this chapter's fixed-date teaching convention be treated directly as regulatory backtesting P&L. An implementation must identify the applicable rules, model configuration and P&L definitions.

The chapter's date-advanced experiment still holds positions constant and omits expenses and interim cash flows. It is a simplified holding-period P&L simulation, not a complete simulation of a desk's actual trading P&L.

## 0.1.5 Full Revaluation vs Sensitivity Based VaR

A VaR implementation must answer three separate questions: how market scenarios are generated, how scenario values are calculated, and which P&L convention is used. These choices can be combined; a method's name cannot replace their definitions.

| Modeling choice | Question to specify | Typical approaches |
|---|---|---|
| Scenario generation | How do market factors move jointly? | Historical simulation, Monte Carlo |
| Scenario valuation | How does a market state map to value? | Full revaluation, sensitivity approximation |
| P&L convention | Does the date advance, and which cash flows enter? | Fixed-date market shocks, holding-period P&L |

Full revaluation[23] evaluates the pricing model at the chosen scenario state. Sensitivity-based valuation approximates the price change using derivatives of that model. Either approach can use an advanced or frozen date, provided its time terms are defined consistently. Monte Carlo specifies a scenario-generation mechanism; its name alone does not imply a reduction in option maturity.

### Full Revaluation

Under this chapter's baseline fixed-date convention:

$$V^{(j)}_{\text{frozen}} = f_t(X^{(j)}_{t+h}; \theta), \qquad L_j = V_t - V^{(j)}_{\text{frozen}}.$$

The pricing model is called again, preserving its nonlinear behavior. Remaining maturity stays at 30/365, with no time decay. Full revaluation can equally use a forward valuation date:

$$V^{(j)}_{t+h} = f(t+h, X^{(j)}_{t+h}; \theta), \qquad L_j = V_t - V^{(j)}_{t+h}.$$

[23] **Full revaluation**: Repricing each instrument under each scenario using the specified pricing model, instead of approximating value changes with local sensitivities. The valuation date is an additional choice.

Remaining maturity then changes from 30/365 to 29/365, bringing time effects into P&L. Both formulas are full revaluation with different time conventions. Repricing preserves the specified model's nonlinearities; it does not remove the pricing model's limitations.

### Sensitivity-Based Valuation

A fixed-date Taylor approximation is:

$$f_t(X_t + \Delta X) - f_t(X_t) \approx g_t^{\mathsf{T}} \Delta X + \tfrac{1}{2} \Delta X^{\mathsf{T}} H_t \Delta X,$$

where $g_t$ is the gradient and $H_t$ the factor Hessian. A complete Hessian includes cross-factor terms; the call's earlier decomposition retains spot Gamma but omits several other second derivatives. In desk terminology, "Delta-Gamma VaR" may refer to a narrower subset of sensitivities, so the included factors and terms must always be stated. The experiment above explicitly adds Vega and Rho.

Time is absent from the fixed-date expression. One may extend it with $\Theta h$, and, where necessary, higher-order and cross-time terms. Sensitivity methods therefore need not exclude Theta, just as full revaluation need not include it when its date is frozen.

| Valuation method | Fixed-date market-shock convention | Date-advanced holding-period convention |
|---|---|---|
| Full Revaluation | Pure time decay excluded | Repricing at the future date includes time effects |
| Sensitivity / Delta-Gamma | No time term; specify factors and Greeks | Add Theta explicitly, with higher-order and cross-time terms where needed |

Historical simulation and Monte Carlo can both generate market scenarios for either time convention. Risk Factor Shock Analysis must likewise specify whether it changes market inputs alone and how it treats the date.

### Implementation Example: Explicit Time Configuration

Open Source Risk Engine (ORE) exposes a separate `includeTheta` option in its historical scenario generator. The `HistoricalScenarioGenerator` constructor defaults to `false` in the inspected revision; the implementation selects either the current date or a date advanced by the risk horizon using the calendar. See Open Source Risk Engine, the pinned source revision in the references, header lines 92–93 and implementation lines 148–150.

This provides a verifiable example of a fixed-date market-shock convention and also demonstrates an alternative time configuration. A constructor default does not represent all analytics, production settings or bank practices. In particular, future-time credit exposure simulation must not be equated with market-risk VaR. Professional documentation records the options actually used rather than substituting a system name for a P&L definition.

### Selecting and Governing the Method

Full revaluation is useful when nonlinearities, large shocks or short remaining maturities make local expansions unreliable. It can be costly for large portfolios or instruments whose pricing requires numerical procedures. Sensitivities offer efficient scenario evaluation and interpretable exposures, but require refreshed Greeks and validation against full repricing over relevant market conditions.

The choice should follow the measurement purpose and numerical evidence, not the method's name. A controlled comparison uses the same positions and scenarios, checks differences in both ordinary and tail outcomes, and reports the effect of advancing time separately. The four rows in the option experiment provide such a comparison. Historical simulation and Monte Carlo are ways to construct scenarios; either can be combined with full revaluation or sensitivity approximations.

Chapter Synthesis

**Risk Factors → Pricing → Revaluation → P&L → Risk Metrics** is the core logic of Market Risk Quant. VaR and ES are outputs of that process. Their meaning is determined by the assumptions upstream: what changes, what is held fixed, how each scenario is valued, and which losses the institution intends to measure.

## Further Topics

- Market Risk Definition and Scope
- Interest Rate Risk
- Equity Risk
- FX Risk
- Commodity Risk
- Credit Spread Risk
- Volatility Risk

## 0.1.6 From Risk Factor Scenarios to Portfolio Loss Distribution

The option case translates market shocks into losses. This section explains how those shocks enter a risk system. A complete risk engine has two connected modeling layers: **scenario generation asks how markets might change and with what probabilities**; **valuation asks what the current portfolio is worth in a given market state**. The layers determine the probability structure and monetary amounts of the loss distribution respectively.

### Scenario Generation: Historical Changes and Conditional Joint Laws

The scenario layer uses $\mathsf{P}_{t,h}(\cdot \mid \mathcal{F}_t)$ to produce joint states $X_{t+h}^{(j)} \in \mathcal{X}$ and weights $w_j \geq 0$, with $\sum_j w_j = 1$. Ordinary Monte Carlo uses $w_j = 1/N$; weighted historical simulation can give greater weight to recent observations. **Monte Carlo is a sampling method, rather than the only way to generate scenarios or a substitute for a factor model.**

**Historical simulation** applies whole vectors of changes from the same historical date to today's market. For positive factors admitting log transformations, one can use:

$$Y_j = \log X_{u_j+h} - \log X_{u_j}, \qquad X^{(j)}_{t+h} = X_t \odot \exp(Y_j).$$

Logarithms, exponentials and multiplication operate componentwise; $\odot$ denotes the componentwise product. Rates can use additive shocks, while curves and surfaces require reconstruction. Keeping date-aligned vectors preserves dependence in the sample; independently shuffling factor dates does not. Historical simulation avoids specifying a normal distribution but still depends on the window, weights, horizon transformations and the representativeness of history.

**Parametric Monte Carlo** specifies a joint law for factor changes $Y$ in suitable coordinates and samples from it. A simplified conditional Gaussian model is:

$$\begin{aligned} Y_j &= \mu_{t,h} + B_{t,h} Z_j, \qquad Z_j \sim \mathcal{N}(0, I), \\ B_{t,h} B_{t,h}^{\mathsf{T}} &= \Sigma_{t,h}, \qquad X^{(j)}_{t+h} = \mathcal{R}(X_t, Y_j). \end{aligned}$$

$\Sigma_{t,h}$ is the conditional covariance of changes; **Cholesky Decomposition** supplies $B_{t,h}$ when it is positive definite. The reconstruction map $\mathcal{R}$ converts log returns, rate changes and other coordinates into valid market inputs. This is equivalent to the earlier example's correlation decomposition followed by factor-specific scaling. Gaussian correlation alone does not adequately represent crisis tail dependence.

Richer models address different tasks: GARCH updates marginal conditional volatility; DCC-GARCH updates correlation; copulas connect marginal distributions; HMM introduces regime transitions. A Transformer or conditional Diffusion model can learn a high-dimensional conditional generation law, subject to validation of horizon consistency, tail coverage and curve/surface validity. A conditional mean forecast is not a scenario distribution, and a large generated sample does not establish model reliability.

## The Interface Between Scenarios and Pricing

Under the baseline fixed-date convention, valuation computes:

$$V_j = f_t(X^{(j)}_{t+h}; \theta), \qquad L_j = V_t - V_j, \qquad \widehat{F}_L(\ell) = \sum_j w_j \mathbf{1}\{L_j \le \ell\}.$$

The loss distribution is the scenario distribution transformed through valuation. For the European call, $C_j = C(S_j, \sigma_j, r_j, \tau_t; K)$. The state $S_j = 95$, $\sigma_j = 25\%$, $r_j = 2\%$ produces $C_j = 1.009287$ and a loss of 1.359048 relative to today's 2.368335. Given consistent inputs and conventions, valuation is the same whether the state comes from historical changes, parametric sampling or a conditional generative model. Probabilities and weights come from the scenario layer.

| Scenario-layer choice | Example valuation engines | Interface requirement |
|---|---|---|
| Historical / parametric Monte Carlo | Black-Scholes for European calls; richer Heston valuation | Scenario quotes reconstruct the required surface and support calibration |
| GARCH marginals + copula sampling | Analytic, PDE or Monte Carlo pricing of options or rate products | Required factors are present; a rate model such as Hull-White needs matching curves and calibration inputs |
| Conditional Transformer / Diffusion | Validated pricing proxies or neural surrogates | Inputs remain within validated coverage and satisfy market and contract constraints |

**Modular separation does not imply statistical independence or unrestricted interchangeability.** Both layers must share factor definitions, quote units, curve/surface reconstruction, calibration rules and contract states. A GARCH variance of historical returns cannot directly replace option implied volatility. Freezing or advancing the valuation date remains a separate P&L convention.

The following core interface accepts `price(state)`, returning the value of the same portfolio in the same reporting currency. Scenario generation and risk-metric aggregation remain outside the function. Replacing the callback changes the valuation method; the callback explicitly holds the time configuration for a frozen or advanced date.

**Python**

```python
def scenario_losses(states, price, base_value):
    return np.array([base_value - price(x) for x
    ↪  in states])
```

**C++**

```cpp
#include <vector>

template<class State, class Pricer>
std::vector<double> scenario_losses(
    const std::vector<State>& states,
    Pricer price, double base_value) {
    std::vector<double> losses;
    losses.reserve(states.size());
    for (const auto& x : states)
        losses.push_back(base_value - price(x));
    return losses;
}
```

## Two Uses of Monte Carlo: Risk Simulation and Pricing

**Risk Monte Carlo** describes future market changes under the real-world measure $\mathbb{P}$, producing horizon-$h$ scenarios for loss forecasting. **Pricing Monte Carlo** estimates the discounted expectation of contractual payments under the risk-neutral measure $\mathbb{Q}$ associated with the chosen numeraire, producing a price at the given scenario state. For a simplified cash payoff $H$ and deterministic rate:

$$v(u, x) = e^{-r(t_{\exp}-u)}\mathbb{E}^{\mathbb{Q}}[H \mid \text{state at valuation time } u \text{ is } x].$$

With stochastic rates, discounting belongs inside the expectation. The contractual payoff law and the market-state law at horizon $h$ have different conditioning, horizons and purposes.

| Comparison | Risk Monte Carlo | Pricing Monte Carlo |
|---|---|---|
| Main question | How are future states and portfolio losses distributed? | What is the present value of payments in this state? |
| Probability measure | Real-world $\mathbb{P}$ | Pricing model's $\mathbb{Q}$ |
| Simulation interval | Risk horizon $h$ | Selected valuation date to remaining payment dates |
| Main output | Joint market scenarios and loss samples | Price estimate in one market state |

**Monte Carlo VaR does not imply Monte Carlo pricing.** This chapter randomly generates spot, implied volatility and rates in the outer layer, then calls the Black-Scholes formula without additional pricing paths. Conversely, an exotic option under historical scenarios may require Monte Carlo pricing. The risk layer simulates future implied-volatility quotes; the pricing layer uses those quotes to calibrate its scenario model. A shared parameter name does not make their dynamics interchangeable.

## Nested Monte Carlo for Exotic Portfolios

A path-dependent contract such as an arithmetic-average Asian option may require an inner pricing simulation for each outer market scenario: **Nested Monte Carlo**[24]. Let $u$ be the selected valuation date: $u = t$ when frozen and $u = t + h$ when advanced. For scenario $j$:

$$\hat{v}_j = \frac{1}{M} \sum_{k=1}^{M} D_{u,t_{\text{exp}}}^{(j,k)} \max(\overline{S}^{(j,k)} - K, 0),$$

$$\overline{S}^{(j,k)} = \frac{A_u + \sum_{\ell=1}^{n_{\text{rem}}} S_{u_\ell}^{(j,k)}}{n_{\text{obs}} + n_{\text{rem}}}.$$

$A_u$ is the sum of observed fixings; $n_{\text{obs}}$ and $n_{\text{rem}}$ count observed and remaining fixings, and $D$ is the discount factor. The outer layer generates states under $\mathbb{P}$; the inner layer generates remaining fixing paths under the scenario-calibrated $\mathbb{Q}$ model. If advancing the date crosses a fixing, the outer state must also supply or simulate that new observation. Horizon-end spot alone is insufficient.

With $N$ outer scenarios, $m$ simulation-priced contracts, $M$ inner paths per contract and $J$ time steps, naive cost is approximately $O(NmMJ)$. For $N = 10{,}000$, $m = 1{,}000$ and $M = 5{,}000$, this already means $5 \times 10^{10}$ contract-pricing paths before counting their time steps. **Inner sampling noise can also perturb loss ordering and tail averages**; it must not automatically be interpreted as market risk.

Production engines combine batching, caching shared curves and calibrations, path reuse and variance reduction such as control variates. Appropriate products can also use Greeks, regression proxies or neural pricing surrogates. A proxy must cover tail shocks, remaining maturities and path states, with full-pricing fallback near boundaries or outside validated coverage. Validation examines price residuals, tail ordering and VaR/ES bias; convergence checks separate outer sampling error from inner pricing noise.

[24] **Nested Monte Carlo**: Outer simulation samples market-risk scenarios; inner simulation estimates conditional contract values in each scenario. The layers commonly use real-world and risk-neutral measures respectively, with different simulated objects and horizons.

### From Market Data to Explainable Risk Numbers

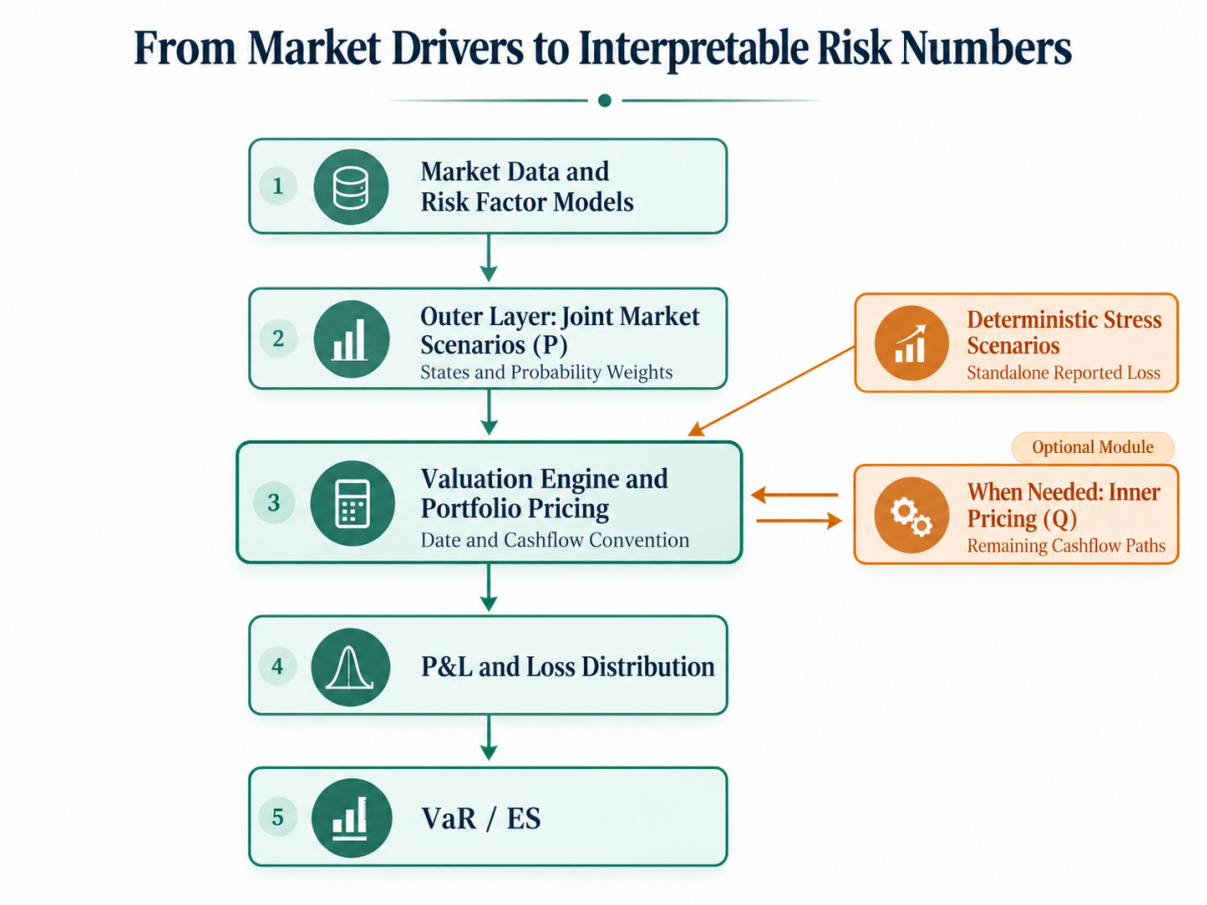



Figure 5: The two-layer risk engine. The outer layer generates market scenarios; valuation turns each state into a value and uses inner Monte Carlo only when pricing requires it. Stress scenarios share valuation, but do not directly yield VaR/ES without probabilities.

The complete workflow is **Market Data → Risk Factor Model → Scenario Generation → Valuation Engine and Portfolio Revaluation → P&L / Loss Distribution → VaR / ES**. Data supplies the common snapshot; factor models define marginal dynamics and dependence; scenarios supply states and probabilities; valuation supplies amounts; P&L conventions govern time and cash flows; risk metrics aggregate the tail.

Stress testing can reuse the valuation interface, but deterministic stress scenarios usually lack defensible occurrence probabilities and should be reported as scenario losses. An explainable risk engine traces every number to its **scenario model, pricing model, inputs, P&L conventions and numerical accuracy**.

## 0.1.7 Interview Questions and Suggested Answers

### Basic

1. Which market risk factors typically affect a foreign-currency corporate bond? Explain why the reporting currency matters.

**Suggested Answer**
Typical factors include interest-rate curves in the bond's currency, the issuer's credit spread, and the exchange rate into the reporting currency. Floating-rate or callable structures can add curve or volatility exposures. For a USD bond reported in CNY, $V^{\text{CNY}} = E_{\text{CNY/USD}} V^{\text{USD}}$: both the bond price and currency conversion must use scenario inputs. The reporting currency determines the conversion exposure. Spread changes and actual default events require distinct treatment.

2. What is a market risk factor?

**Suggested Answer**
A market risk factor describes a market state and drives instrument value or the distribution of future P&L. Examples include spot prices, exchange rates, curve nodes, credit spreads, and implied volatility. A bond is an instrument; its rates and spreads are valuation drivers. Factor definitions must specify units, tenors, and quotation conventions and support consistent scenario revaluation.

3. Why can portfolio value be represented as $V = f(X)$?

**Suggested Answer**
Given positions, contractual terms, and model rules, market inputs map into portfolio value through pricing rules. The fuller representation is $V_t = f(t, X_t; \theta)$, where $t$ controls the valuation clock and $\theta$ contains fixed settings. The function can range from quantity times price to an aggregation of complex pricing models. This representation establishes a modeling framework; it does not guarantee that the selected model captures every factor or prices without error.

4. Which inputs of this call are stochastic market factors, and which are contractual terms or deterministic time states?

**Suggested Answer**
In this scenario model, $S$, $\sigma$, and $r$ are stochastic market factors. Strike $K$, contractual expiry, and position size are fixed terms or settings. Remaining maturity $\tau$ follows from expiry and the valuation date: it stays constant when the date is frozen and decreases with calendar time when the date advances. Although $\sigma$ is a pricing input, it is shocked as an implied-volatility risk factor; stochastic treatment depends on the risk model.

5. Can an option's value change when spot, implied volatility and rates remain unchanged?

**Suggested Answer**
Yes, if the valuation date advances and remaining maturity decreases. In this example, reducing maturity from 30/365 to 29/365 lowers value from 2.368335 to 2.327215, giving a loss of 0.041121 with market inputs unchanged. If the date is also frozen, identical inputs

and model rules produce identical values. Time P&L depends on how the pricing clock is treated.

6. What is recalculated in full revaluation, and what is approximated in a sensitivity-based method?

**Suggested Answer**
Full revaluation reruns pricing models under each scenario to calculate portfolio value. A sensitivity method approximates value changes using initial-state gradients, Hessians, or selected Greeks. A fair comparison requires consistent factors and P&L definitions. Full revaluation retains the specified model's nonlinear behavior but remains subject to that model's limitations and input quality.

## Intermediate

7. In Example 0.1, how does doubling the position size change the P&L distribution? Does the probability of loss change?

**Suggested Answer**
With fixed positions and no nonlinear size effects such as transaction costs, $\Delta_h V_t' = 2\Delta_h V_t$ and $L' = 2L$. The two outcomes become +4,000 CNY and −6,000 CNY, retaining probabilities of 60% and 40%. The probability of loss is unchanged, while VaR and ES double under positive scaling. Loss amounts and the probability of loss describe different aspects of risk.

8. Explain the process from risk factor movement to portfolio loss distribution.

**Suggested Answer**
Specify positions, reporting currency, horizon, and the P&L convention. Define a conditional joint distribution $\mathsf{P}_{t,h}(\cdot \mid \mathcal{F}_t)$ on the factor space $\mathcal{X}$ and generate scenarios. Revalue each under consistent rules and compute $(\Delta_h V_t)^{(j)}$ and $L_j = -(\Delta_h V_t)^{(j)}$. Losses and their probabilities form the loss distribution. Extract VaR and ES, interpreting them alongside stress results and model validation. The chain is Risk Factors → Scenario Generation → Pricing / Revaluation → P&L → Risk Metrics.

9. Why do we define $L = -\Delta_h V_t$ in risk management?

**Suggested Answer**
This convention makes losses positive and places severe losses in the right tail, making loss quantiles and tail averages straightforward to define. It changes reporting direction, not economic outcomes. Gains correspond to $L < 0$ and must remain in the distribution. Replacing $L$ by $\max(L, 0)$ changes the distribution and can change risk measures.

10. How would you decompose option P&L into Delta, Gamma, Vega and Theta effects?

**Suggested Answer**
Evaluate Greeks at today's state using consistent decimal and year-fraction units:

$$\Delta_h V_t \approx \delta\,\Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2 + \nu\,\Delta\sigma + \rho\,\Delta r + \Theta h.$$

These terms represent Delta, Gamma, Vega, Rho, and time effects. Omit $\Theta h$ when the valuation date is frozen. This local expansion excludes cross and higher-order terms such as Vanna and Volga. Report the difference from exact revaluation as a residual; do not automatically assign it to Theta.

11. Why does a rise in implied volatility not guarantee a profit for the long call in this scenario?

**Suggested Answer**
Higher implied volatility benefits the long call through Vega, but the simultaneous fall in spot from 100 to 95 has a larger adverse effect. Fixed-date full revaluation reduces value from 2.368335 to 1.009287, giving a loss of 1.359048 without time decay. Advancing the date by one day raises the loss in this example to 1.396950. A favorable move in one factor cannot determine total P&L under a joint scenario.

12. A trader says "the market did not move, so the option should have zero P&L change." How would you respond?

**Suggested Answer**
Clarify whether the trader means fixed-date market-shock P&L or elapsed-time holding-period P&L. The former is zero when all market inputs are unchanged; the latter can still contain Theta, carry, and relevant cash flows. This long call loses approximately 0.041121 when the date advances one day without a market move. Unchanged market quotes do not imply unchanged inputs to the entire valuation calculation.

13. Explain the difference between full revaluation VaR and delta-gamma VaR.

**Suggested Answer**
With identical scenarios, positions, and time conventions, full revaluation calls the pricing model in each scenario. Delta-gamma uses a local first- and second-order expansion, usually reducing cost but potentially losing accuracy under large shocks or strong nonlinearities. Specify whether the approximation includes Vega, Rho, and cross-factor terms. Either method can freeze or advance the date; Theta inclusion is a separate convention. Explain VaR differences through comparable scenario losses.

14. How do risk Monte Carlo and pricing Monte Carlo differ?

**Suggested Answer**
Risk Monte Carlo samples joint horizon-end market states under the real-world measure $\mathbb{P}$, then revalues them to produce losses. Pricing Monte Carlo estimates discounted remaining payments under $\mathbb{Q}$ conditional on a given market and contract state. Measures, horizons, simulated objects and outputs differ. This chapter combines random outer scenarios with Black-Scholes pricing: Monte Carlo VaR does not require Monte Carlo pricing.

15. Can scenarios be generated without Monte Carlo? Is historical simulation assumption-free?

**Suggested Answer**
Yes: historical simulation applies date-aligned joint factor changes to current states; deterministic stress scenarios can also be designed separately. Historical simulation avoids

imposing normality but assumes choices of window, weights, horizon and shock transformations, and historical representativeness. GARCH, copulas and conditional generative models specify dynamics or distributions; sampling is an implementation step. Stress losses without probabilities do not directly aggregate into VaR/ES.

## Advanced

16. Why are a best point forecast of a future stock price and the distribution of future position losses different modeling objectives?

**Suggested Answer**
A point forecast gives one future price, often summarizing a mean or another central location. A loss distribution also describes outcome probabilities, tails, and their transformation through valuation. Models with the same point forecast can have very different tails. For nonlinear instruments, generally $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$, so pricing at a forecast market state cannot replace a distribution of scenario losses.

17. What can go wrong when marginal factor scenarios are generated independently for a portfolio with offsetting exposures?

**Suggested Answer**
Independent generation changes the joint distribution even if every marginal distribution is correct. It can distort hedges or omit simultaneous extremes. For linear exposure $\Delta_h V_t = a\Delta X_1 + b\Delta X_2$, the variance contains $2ab\,\mathrm{Cov}(\Delta X_1, \Delta X_2)$; imposing independence removes this term. Risk can be overstated or understated depending on exposure signs and actual dependence. Joint simulation or resampling whole date-aligned factor vectors preserves the modeled dependence.

18. How should empirical ES be calculated when several scenarios lie exactly at the VaR threshold?

**Suggested Answer**
**VaR identifies a loss quantile; ES measures the average loss within the worst $1-\alpha$ probability mass.** For a discrete distribution, averaging all scenarios with $L > \mathrm{VaR}_\alpha$, or all with $L \geq \mathrm{VaR}_\alpha$, may select the wrong tail probability.
**The quantile and its boundary mass.** Let $F_L$ be the loss distribution function. Under this chapter's inverse-CDF convention:

$$v = \mathrm{VaR}_\alpha(L) = \inf\{\ell : F_L(\ell) \geq \alpha\}, \qquad \Pr(L < v) \leq \alpha \leq \Pr(L \leq v).$$

A 99% VaR of 100 therefore means that losses do not exceed 100 with probability of at least 99%. It does not guarantee that exactly 1% of losses strictly exceed 100. When the threshold has probability mass, the tail may need to include part of that mass.

**Case: 1,000 equally weighted scenarios.** Suppose losses, expressed in a common currency unit, have the following empirical distribution:

| Loss $L$ | Scenario count | Empirical probability | Tail inclusion mass |
|---:|---:|---:|---:|
| 50 | 985 | 98.5% | 0 |
| 100 | 12 | 1.2% | 7 |
| 150 | 1 | 0.1% | 1 |
| 200 | 1 | 0.1% | 1 |
| 500 | 1 | 0.1% | 1 |

Here $F_L(100^-) = 98.5\%$ and $F_L(100) = 99.7\%$, so $\mathrm{VaR}_{99\%} = 100$. The target tail contains $1{,}000 \times 1\% = 10$ scenarios' worth of probability. The three losses strictly above 100 account for only 0.3%; another 0.7%, equivalent to seven scenarios, must come from the twelve losses equal to 100. Since their losses are identical, an equivalent allocation assigns each of those twelve scenarios a tail inclusion weight of $7/12$.

$$\mathrm{ES}_{99\%} = \frac{7 \times 100 + 150 + 200 + 500}{10} = 155.$$

Averaging only $L > 100$ gives $850/3 \approx 283.33$, covering only the worst 0.3%. Averaging all $L \geq 100$ gives $2{,}050/15 \approx 136.67$, covering the worst 1.5%. Neither is the required 1% tail mean.

**How the excess-loss formula handles the boundary.** For integrable losses, with $(x)^+ = \max(x, 0)$:

$$\mathrm{ES}_\alpha(L) = v + \frac{\mathbb{E}[(L - v)^+]}{1 - \alpha}.$$

The expectation $\mathbb{E}[(L - v)^+]$ averages excess losses over **all scenarios**, rather than conditioning on threshold exceedance. In this case:

$$\hat{\mathbb{E}}[(L - 100)^+] = \frac{(150 - 100) + (200 - 100) + (500 - 100)}{1{,}000} = 0.55,$$

$$\mathrm{ES}_{99\%} = 100 + \frac{0.55}{0.01} = 155.$$

The formula assigns the threshold loss of 100 to the entire 1% tail, then adds the excess losses from strict exceedances. It therefore retains the required boundary mass automatically. In general, include all probability above $v$, then add $(1 - \alpha) - \Pr(L > v)$ from the mass at $v$. For a continuous distribution with no atom at the threshold and $F_L(v) = \alpha$, the expression reduces to $\mathrm{ES}_\alpha = \mathbb{E}[L \mid L > v]$.

**Fractional tails and risk interpretation.** If confidence in this case is changed to $\alpha = 99.25\%$, the target tail has 7.5 scenarios' worth of mass and $v$ remains 100. Include the three exceedances and 4.5 scenarios' worth of threshold mass, giving $(850 + 4.5 \times 100)/7.5 \approx 173.33$. Even without ties, fractional boundary weights are needed whenever $N(1 - \alpha)$ is not an integer.

If only the extreme loss of 500 in the table is replaced by 1,500, 99% VaR remains 100, while ES rises from 155 to 255. Thus **VaR describes the tail threshold, while ES also reflects loss severity within that tail**. Historical simulation and Monte Carlo implementations must average exactly the specified tail probability mass.

19. Why does a one-day risk horizon not necessarily reduce an option's remaining maturity by one day? Does holding positions constant imply a frozen valuation date?

**Suggested Answer**
The risk horizon sets the duration of simulated market changes; the valuation date sets the pricing clock. One-day factor changes can be applied to $f_t$ with maturity unchanged, or valued using $f(t+h, X_{t+h})$ with an advanced date. Static positions hold quantities and contracts fixed, not the valuation clock. These choices must be documented separately.

20. Why does the Greek approximation differ from full revaluation, and what does the residual represent?

**Suggested Answer**
Greeks are local derivatives at the initial state. A finite-order expansion cannot reproduce every nonlinear response; large joint shocks, surface changes, and short maturity can increase approximation error. In this controlled example, residuals mainly reflect omitted cross terms, higher-order terms, and, when time advances, higher-order time effects. In production, input alignment, pricing sources, and numerical errors also require investigation. A residual alone neither proves model failure nor identifies one particular Greek.

21. Why must the scenario distribution be distinguished from the risk-neutral measure used for pricing?

**Suggested Answer**
Risk scenarios generally describe future market probabilities under the real-world measure $\mathbb{P}$. Pricing models use a risk-neutral measure $\mathbb{Q}$ to calculate model-consistent discounted expected payoffs. Their purposes differ. A risk engine can generate states under $\mathbb{P}$ and invoke a pricing model using $\mathbb{Q}$ at each state. Risk-neutral pricing does not justify directly using its drift or scenario probabilities to forecast actual losses.

22. For an option portfolio, should time decay be included in VaR calculation?

**Suggested Answer**
The measurement objective and applicable P&L definition determine the treatment. Fixed-date market-shock losses exclude pure time decay by construction. Future-date holding-period P&L advances time and treats interim cash flows consistently. Both full revaluation and sensitivity approximations can implement either convention. Backtesting and regulatory comparisons must additionally follow the applicable APL, HPL, and RTPL rules; labels such as Monte Carlo or bank VaR do not settle the issue.

23. Why is the exact time-roll P&L under a shocked market state not necessarily equal to today's Theta multiplied by the horizon?

**Suggested Answer**
**Theta is a function of time and market state, not a fixed constant.** For an option with fixed contractual expiry, remaining maturity decreases as calendar time advances. Its time sensitivity is defined as:

$$\Theta(t, X) = \partial_t f(t, X).$$

It measures the instantaneous effect of time passage with market inputs fixed. Today's

$\Theta(t, X_t)$ generally differs from the shocked-state, unadvanced-date $\Theta(t, X_{t+h})$ because spot, volatility, and rates have changed. Even at a fixed shocked state, Theta changes as remaining maturity decreases. There are therefore two distinctions: **the market state at which the derivative is evaluated, and changes in that derivative over a finite interval.** First apply the market shock at date $t$, then hold $X_{t+h}$ fixed while advancing the valuation clock. If the interval contains no expiry, payment, or other value jump, and valuation is sufficiently smooth in time, exact roll satisfies:

$$\begin{aligned} f(t+h, X_{t+h}) - f(t, X_{t+h}) &= \int_t^{t+h} \Theta(u, X_{t+h})\, du, \\ &\approx \Theta(t, X_{t+h})h. \end{aligned}$$

The integral accumulates instantaneous Theta at that fixed market state. The last step is a good approximation only over a short interval with little variation in Theta. **This continuous-time identity also explains discrete one-day scenarios.** It follows from the fundamental theorem of calculus: $X_{t+h}$ is fixed throughout the integral, rather than evolving along a stochastic market path as in an Itô decomposition. Substituting today's $\Theta(t, X_t)h$ additionally requires controlling the effect of the market-state change.

For this chapter's option, $h = 1/365$ and Theta is annualized. Today's Theta times the horizon is approximately $-0.040797$; the shocked-state Theta times the horizon is $-0.037762$; exact one-day roll P&L is $-0.037902$. The first difference reflects the changed market state; the second reflects Theta's variation during the roll.

**Whether VaR includes time roll still depends on the P&L convention.** A frozen valuation date omits the pure time term. An advanced date permits scenario-by-scenario repricing or an explicitly stated Theta approximation with its error. Crossing expiry or settlement also requires contractual cash flows and the resulting position changes; pre-expiry Theta cannot be extrapolated beyond contract termination.

24. Why do static HPL positions not imply a frozen valuation date? How does daily P&L Explain differ from the regulatory PLA test?

**Suggested Answer**

**Static positions specify the trade set; the valuation date specifies the pricing clock. These are separate dimensions.** HPL asks what the previous closing portfolio would earn or lose if held until today. It excludes intraday trading and new or modified deals, while updated market data and shorter remaining maturity still affect value. Let $P_t$ denote previous closing positions. In a simplified case without intervening cash flows or valuation adjustments:

$$\mathrm{HPL}_{t,h} = f(t+h, X_{t+h}; P_t) - f(t, X_t; P_t).$$

Both valuations use $P_t$, but the date advances from $t$ to $t+h$. For example, the chapter's call moves from 30 to 29 days remaining and produces approximately $-0.041121$ P&L per unit even with spot, volatility, and rates unchanged. Frozen-date market-shock repricing gives zero in those conditions, so it cannot directly replace this holding-period P&L. Payments, expiry, or settlement additionally require cash-flow and contract-state reconciliation.

**The objects represented by APL, HPL, and RTPL.**

| Series | Positions and valuation scope | Principal regulatory use |
|---|---|---|
| **APL: Actual P&L** | Reflects actual trading activity, including intraday and new or modified deals; uses daily P&L pricing with regulatory adjustments to included items. | VaR backtesting. |
| **HPL: Hypothetical P&L** | Revalues previous closing positions using daily P&L pricing and current market data. | VaR backtesting; the comparator for PLA. |
| **RTPL: Risk-Theoretical P&L** | Uses the same static positions, with the risk model's factors and valuation engine. | Comparison with HPL to assess risk-model coverage. |

**Regulatory APL is not unadjusted accounting P&L.** MAR32.26 excludes fees, commissions, and specified valuation adjustments from both APL and HPL. All realized income and expenses therefore cannot simply be copied into regulatory APL. MAR32.27 includes passage-of-time P&L in APL and requires consistent time treatment in HPL and RTPL, including applicable Theta, carry, and funding costs; RTPL's time component cannot be arbitrarily set to zero. Nor is RTPL merely a Delta/Gamma/Vega attribution table: its engine can capture nonlinear pricing, and MAR32.22 requires its factors to include regulatory ES-model factors and the corresponding non-modellable risk factors.

**Daily P&L Explain asks why the desk made or lost money today.** It reconciles a specified P&L measure to sources such as Delta, Gamma, Vega, Theta, carry, trading activity, and residuals. For example, in millions, $-3-1+0.5-1-0.5=-5$ can represent Delta, Gamma, Vega, Theta, and residual contributions respectively. Residuals require investigation rather than automatic assignment to a Greek. This is a P&L explanation process.

**The regulatory PLA (P&L Attribution) test asks whether the risk model captures the material drivers of desk P&L.** It compares RTPL with HPL on the same static positions, removing intraday trading effects and assessing differences in factor coverage and valuation engines. MAR32.34–32.35 use the latest 250 trading days: **Spearman rank correlation** measures joint variation, while the **Kolmogorov–Smirnov (KS) statistic** measures distributional differences. A day with HPL of $-10$ million and RTPL of $-2$ million has an 8 million discrepancy, prompting investigation of missing factors, nonlinear valuation, or data alignment. That single discrepancy does not establish a PLA failure.

**VaR backtesting asks whether observed outcomes support the predicted loss quantile.** MAR32.4 compares one-day VaR separately with APL and HPL; HPL for PLA must be identical to HPL for backtesting. If VaR $= 10$ million and the corresponding observed P&L is $-12$ million, loss $L = 12 > 10$ constitutes an exception. Backtesting assesses quantile forecasts, PLA assesses agreement between two P&L series, and daily P&L Explain identifies sources of P&L.

These conventions follow Basel MAR32, paragraphs 32.4, 32.22–32.29, and 32.34–32.35; implementation must follow the applicable jurisdiction's rules.

25. Design a comparison that separates approximation error from the effect of advancing the valuation date.

**Suggested Answer**

**Treat valuation method and time convention as two independent experimental dimensions.** Full revaluation is a benchmark under the same pricing model. It avoids truncating a local Greek expansion but remains subject to pricing assumptions and numerical accuracy; it is not an error-free true value.

**Control inputs and sampling first.** Use identical positions, contracts, market snapshots, factor representations, scenario weights, calendars, and cash-flow rules. For Monte Carlo, generate and save joint scenarios $X_{t+h}^{(j)}$ once, then reuse each scenario in all four calculations. This is a paired **Common Random Numbers** design. The same seed alone may produce different scenarios when random-number consumption differs. If pricing itself uses inner simulation, also control its random draws and convergence tolerances, or estimate its numerical noise.

Measure all P&L against the same initial value $V_t$. For the chapter's option, explicitly retain today's Delta, Gamma, Vega, and Rho, adding today's Theta in the advanced-date approximation:

$$\begin{aligned} A_j &= f(t, X_{t+h}^{(j)}) - V_t, \\ B_j &= f(t+h, X_{t+h}^{(j)}) - V_t, \\ C_j &= \delta\,\Delta S_j + \tfrac{1}{2}\Gamma(\Delta S_j)^2 + \nu\,\Delta\sigma_j + \rho\,\Delta r_j, \\ D_j &= C_j + \Theta(t, X_t)h. \end{aligned}$$

The Delta term must include its sensitivity coefficient $\delta$. This specification retains spot Gamma but omits Vanna, Volga, higher-order market terms, and market-time cross terms. Other sensitivity schemes must state their included terms separately.

**The 2×2 experiment and the chapter's stress scenario.** Amounts below are P&L per unit, using $S : 100 \to 95$, $\sigma : 20\% \to 25\%$, unchanged rates, and $h = 1/365$.

| Valuation method | Frozen date $t$ | Advanced date $t+h$ |
|---|---|---|
| **Full revaluation** | $A_j$; example $-1.359048$ | $B_j$; example $-1.396950$ |
| **Sensitivity approximation** | $C_j$; example $-1.175099$ | $D_j$; example $-1.215896$ |

**Compare methods within a time convention, and time conventions within a method.** $A_j - C_j$ and $B_j - D_j$ measure benchmark residuals with frozen and advanced dates respectively. $B_j - A_j$ is exact roll at the shocked state; $D_j - C_j$ is the approximation's added time term, which need not equal the exact Theta/carry effect. A comparison across both dimensions can be decomposed, with an interaction also checked:

$$\begin{aligned} B_j - C_j &= (A_j - C_j) + (B_j - A_j), \\ (B_j - D_j) - (A_j - C_j) &= (B_j - A_j) - (D_j - C_j). \end{aligned}$$

In this example, $B - C \approx -0.221851$ combines the approximation residual $A - C \approx -0.183948$ and exact roll $B - A \approx -0.037902$. The approximate time term is $D - C \approx -0.040797$, so the methods' time effects differ by approximately 0.002895: the advanced-date residual differs from the frozen-date residual. Differences use unrounded results and cannot all be labelled Delta-Gamma error.

**Validation must cover scenario P&L, tails, and risk metrics.** Pair residuals by scenario identifier and inspect bias, root mean square error, extremes, and regions with large shocks or short maturity. Then calculate VaR/ES for all four loss samples, $L = -P\&L$,

using identical confidence levels, weights, and quantile conventions. Date advancement may change tail membership and ordering. Aggregate metrics measure risk impact but cannot alone identify its source; their differences are not the risk metrics of the residual sample. Model benchmarking requires each difference to be traceable to portfolios, models, inputs, P&L conventions, or numerical accuracy.

26. Can historical simulation use Greeks, and can Monte Carlo use full revaluation? Explain why scenario generation, scenario valuation and time treatment are three separate choices.

**Suggested Answer**
Yes. Historical simulation constructs scenarios from historical factor changes and can price their P&L using Greeks. Monte Carlo draws from a specified joint probability model and can fully revalue each draw. Scenario generation determines how markets move; valuation determines how states map into values; the P&L convention determines date and cash-flow treatment. These choices must be consistent, but none can be inferred from the name of another.

27. What information is missing from each statement: "Full Revaluation VaR necessarily includes Theta" and "all bank VaR freezes time"?

**Suggested Answer**
The first statement mistakes a valuation method for a time convention: full revaluation means rerunning pricing, and a frozen date excludes pure time decay. The second generalizes an implementation or internal convention to every bank, ignoring objectives, system options, and regulatory P&L requirements. Both omit explicit valuation dates, horizons, cash-flow treatment, and comparison conventions. These settings determine whether Theta enters the risk measure.

28. Why separate scenario generation and valuation, and why are combinations constrained?

**Suggested Answer**
The scenario layer supplies market states and probabilities; valuation supplies conditional contract amounts. Separation supports independent module validation and replacement. Both layers must match factor definitions, units, curve/surface reconstruction, calibration and contract states, with explicit date and P&L conventions. This is architectural separation, not statistical independence. GARCH return volatility cannot directly replace pricing implied volatility.

29. How is nested Monte Carlo constructed for an Asian option, and why is horizon-end spot insufficient?

**Suggested Answer**
The outer layer generates market states under $\mathbb{P}$; the inner layer simulates remaining fixing paths under a scenario-calibrated $\mathbb{Q}$ model to estimate discounted payments. Arithmetic-average payoffs also depend on observed fixing sums and counts. If an advanced date crosses a fixing, the outer state must supply the new observation. Identical horizon-end spots can have different realized averages and therefore different prices.

30. Why is nested Monte Carlo costly, and how can inner noise affect VaR/ES?

**Suggested Answer**
Naive cost is $O(NmMJ)$: outer scenarios, contracts, inner paths and time steps multiply. Finite inner samples produce pricing errors that can alter loss rankings, scenarios near the VaR boundary and ES tail averages. Hold outer scenarios fixed while increasing inner accuracy or repeating inner draws, then separately assess outer convergence. The resulting numerical discrepancies must not all be attributed to market risk.

31. How would you accelerate thousands of exotic options and validate the risk engine?

**Suggested Answer**
Batch pricing, cache shared curves and calibrations, reuse paths and reduce variance. Use Greeks or validated regression/neural proxies for suitable contracts, including maturity and path state, with full-pricing fallback outside coverage. On common positions, scenarios and P&L conventions, validate price residuals, tail ordering and VaR/ES; assess approximation error and inner/outer sampling errors separately.